

Exhibit 100

U.S. Patent No. 10,025,731 Domestic Industry Claim Chart (Samsung)

EXHIBIT 100

Domestic Industry Claim Chart for Claims of U.S. Patent No. 10,025,731 (the “’731 Patent”) Based on Samsung Practicing ’731 Products

As shown in the chart below, the Samsung Practicing ’731 Products, which include Samsung DDR5 RDIMMs and MRDIMMs, practice at least claim 1 of the ’731 patent. It should be understood that Netlist’s contentions are not limited to these products but instead encompass any and all reasonably similar products and components that practice the claims in the same or similar manner as discussed below. Netlist may supplement, modify or amend its contentions under the doctrine of equivalents based on any claim constructions, additional information obtained through formal discovery or other means, and/or based on other circumstances including any assertions by Respondents that the Samsung Practicing ’731 Products do not practice the ’731 patent as to any one or more of the limitations in the claim(s) below.

To the extent Respondents contend that any differences are alleged to exist between the asserted claims below and the Samsung Practicing ’731 Products, such differences are insubstantial and the Samsung Practicing ’731 Products perform substantially the same function, in substantially the same way, to yield substantially the same result, and therefore the Samsung Practicing ’731 Products practice the claims below under the doctrine of equivalents. Netlist may supplement, modify or amend its contentions under the doctrine of equivalents based on any claim constructions, additional information obtained through formal discovery or other means, and/or based on other circumstances including any assertions that the Samsung Practicing ’731 Products do not satisfy the technical domestic industry by Respondents as to any one or more of the limitations in the claims below.

U.S. Patent No. 10,025,731 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
[1.a] [i] A memory module [ii] operable to communicate data with a system memory controller via a memory bus [iii] in response to address and control signals from the system memory controller, comprising:	[i] Samsung's DDR5 RDIMMs and MRDIMMs, and all related versions (e.g., VLP versions or versions also with a separate clock driver) are each a memory module. See, e.g., Ex. CC-94 (Samsung Module Handling Guide) at 2: Multiplexed Rank DIMM Registered DIMM For example, "DIMM" stands for dual in-line memory module. As another example, Samsung's web site advertises "[m]emory modules for extensive use," and lists RDIMMs as examples of the memory modules offered by Samsung. See Ex. CC-95 (Samsung DRAM Modules). [ii] Samsung's DDR5 RDIMMs and MRDIMMs are each operable to communicate data (e.g., to read or write data) with a system memory controller (e.g., a memory controller in a host computer's CPU) via a memory bus (e.g., as the pathway between a CPU and DRAMs for data communication, which would go through a memory slot in the case of communication between a memory controller

U.S. Patent No. 10,025,731 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	and a DIMM). For example, according to Ex. CC-67 (DevX Memory Bus Definition) at 2 (emphasis added), the term “Memory Bus” “<i>refers to the data pathway between the central processing unit (CPU) and the computer’s main memory (RAM).</i>” Essentially, it <i>serves as the communication channel that enables these two critical components to exchange information</i>, ensuring the efficient functioning of the computer.” As another example, Samsung’s web site states that “RDIMM (registered memory) <i>attaches a register between the CPU and the DRAM chip for data transmission</i>, which reduces the distance of parallel transmission and improves transmission efficiency. It is a type of memory chip utilized most often in servers. It provides better stability and scalability for systems running on a server by putting a less electrical load on <i>a memory controller.</i>” Ex. CC-144 (Samsung Module RDIMM) at 8 (emphasis added) (FAQs: “What is RDIMM?”). Data communication between a DDR5 memory module and a system memory controller in a CPU is via a memory bus, which would be understood as the data pathways between a CPU and SDRAMs. For instance, the Samsung DDR5 RDIMM Datasheet shows DQ (data) communication pathways between “Host” which includes the memory controller and “DRAM” which are mounted on DIMMs:

Claim 1

Evidence of Use

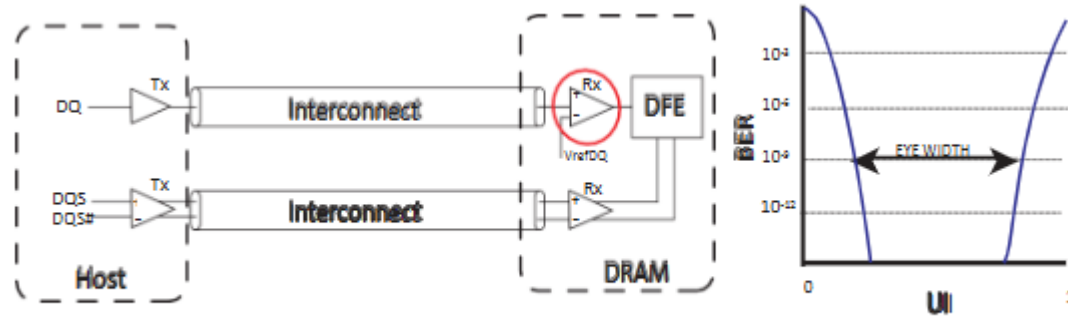


Figure 17. Example of DDR5 Memory Interconnect

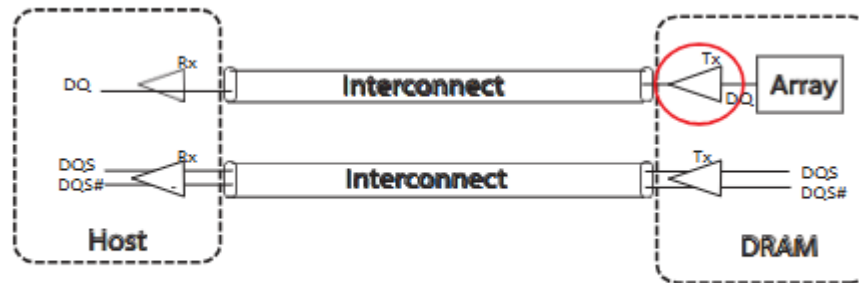


Figure 34. Example of DDR5 Memory Interconnect

Ex. CC-89 (Samsung DDR5 RDIMM Datasheet) at 49, 67.

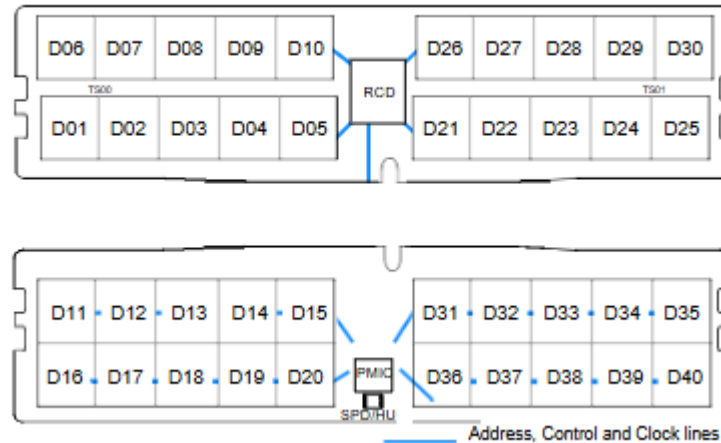
See also, e.g., Ex. CC-47 (DDR Memory & Challenges in PCB Design) at 4 (annotated):

Claim 1	Evidence of Use
	The diagram illustrates a memory hierarchy. At the top is a grey box labeled 'Controller' with a red border. Below it is a grey box labeled 'Cache'. Three arrows point from the 'Controller' box to three separate boxes below: a yellow box labeled 'Cache (SRAM)', a green box labeled 'Primary Memory', and a blue box labeled 'Secondary Memory'. The 'Primary Memory' box contains sub-boxes for 'RAM', 'ROM', 'SRAM', 'PROM', 'SDRAM DDR 1,2,3,4,5' (which has a red border), and 'EEPROM'. The 'Secondary Memory' box contains sub-boxes for 'Magnetic', 'Hard Disk', and 'Flash'. <i>Types of memory</i> [iii] The data communication between a Samsung memory module and a system memory controller is <i>in response to address and control signals</i> (e.g., those associated with a read or write operation) <i>from the system memory controller</i>. For example, as shown in a representative image of a Samsung DDR5 RDIMM, the memory module receives control and address signals from the memory controller (via the edge pins of the memory module) and regenerates module-level control/address signals for outputting to DDR5 SDRAMs' input circuitry for control/address signals:

U.S. Patent No. 10,025,731 Domestic Industry Claim Chart (Samsung)

Claim 1

Evidence of Use



Ex. CC-89 (Samsung DDR5 RDIMM Datasheet) at 33; *see also id.* at 29, 25, 21.

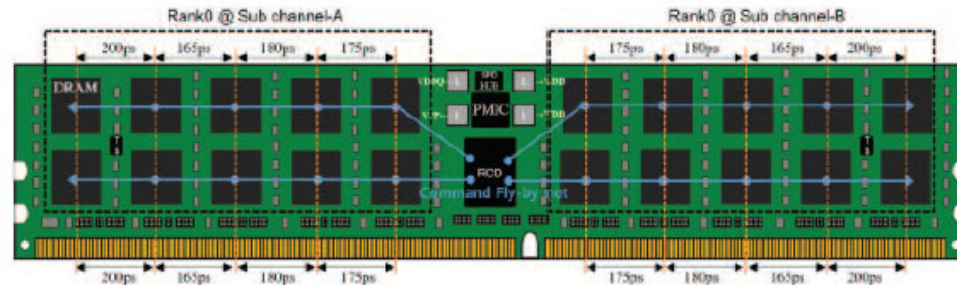


Figure 3. The operation delay due to fly-by topology of command and address nets in DDR5 RDIMM 2Rx4.

Ex. CC-45 (Na 2022) at 931, Fig. 3 (showing RCD outputting control/address signals to SDRAMs).

See also Ex. CC-96 (Rambus DDR5 RCD) at 1 (Rambus is a supplier of Samsung DDR5 RDIMM.)

U.S. Patent No. 10,025,731 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	(“The Rambus DDR5 RCD provides Command/Address (CA) and clocks to the DDR5 memory devices in RDIMMs. In DDR5 DIMM architectures, the module is organized into two independent 40-bit subchannels (32 bits data, 8 bits ECC). Each subchannel is organized as either 10 devices in a x4 configuration with 2 ranks or 5 devices in a x8 configuration with 2 ranks. The RCD distributes C/A signals and clocks for each subchannel, rank and nibble. Sideband channel communication with the CPU or motherboard Baseboard Management Controller (BMC) is enabled through a 12.5 MHz (max.) I3C bus interface.”): <pre> graph LR subgraph RCD direction TB subgraph Channel_A [Channel A] direction TB CA_T[Channel A Transmit] CA_R[Channel A Receive] end subgraph Channel_B [Channel B] direction TB CB_T[Channel B Transmit] CB_R[Channel B Receive] end Logic[Logic] Clocking[Clocking] Regulators[Regulators] I2CI3C[I2C/I3C] end CA_T --> ToDRAM_CA[To DRAM Command/Address] CA_T --> ToDRAM_CS[To DRAM Chip Select] CA_T --> ToDRAM_Clk[To DRAM Clocks] CB_T --> ToDRAM_CA CB_T --> ToDRAM_CS CB_T --> ToDRAM_Clk ToHost_CA[From Host Command/Address] --> CA_R ToHost_CS[From Host Chip Select] --> CA_R ToHost_CB[From Host Command/Address] --> CB_R ToHost_CSC[From Host Chip Select] --> CB_R InputClock[Input Clock] --> Clocking InputPower[Input Power] --> Regulators I2CI3C <--> SidebandBus[Sideband Bus] </pre> DDR5 RCD High-Level Functional Block Diagram Ex. CC-48 (Montage M88DR5RCD01) at 1 (Montage is also an RCD supplier for Samsung’s DDR5 RDIMMs.) (“It is a two-channel RCD designed for driving addresses and controlling nets between the host controller and the DDR5 SDRAMs.”):

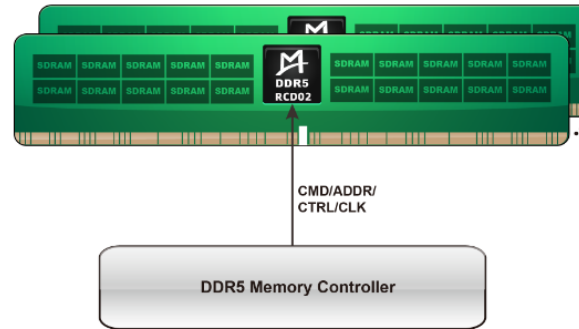
U.S. Patent No. 10,025,731 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	The diagram illustrates a memory module architecture. A green rectangular module contains a grid of components. The top row includes 'PMIC', 'CPU', and 'GPU'. The bottom row includes 'RCD'. The module is connected to a 'DDR5 Memory Controller' via a line labeled 'CMD/ADDR/CTRL/CLK'. Ex. CC-87 (Montage M88DR5RCD02) at 1 (“M88DR5RCD02 is a Gen2 DDR5 Registering Clock Driver (RCD02) with a maximum data rate of 5600MT/s. Compliant with the latest JEDEC DDR5RCD02 specification, the RCD02 contains two separate channels and supports 1.1V VDD supply. It is designed for use on DDR5 RDIMMs to enable memory systems with higher capacity and lower power consumption for data-intensive applications” such as high-performance computing (HPC) and artificial intelligence (AI).); Ex. CC-97 (Montage M88DR5RCD03) at 1 (“M88DR5RCD03 is a Gen3 DDR5 Registering Clock Driver (RCD03) with a maximum data rate of 6400MT/s. Compliant with the latest JEDEC DDR5RCD03 specification, the RCD03 contains two separate channels and supports 1.1V VDD supply. Designed for use on DDR5 RDIMMs, it sets a new bar for DDR5 memory performance in upcoming server platforms and unlocks dramatic capacity, bandwidth, and latency improvements to meet the growing demands of data center and cloud computing applications.”); Ex. CC-98 (Montage M88DR5RCD04) at 1 (“M88DR5RCD04 is a Gen4 DDR5 Registering Clock Driver (RCD04) with a maximum data rate of 7200MT/s. Compliant with JEDEC DDR5RCD04 specification, the RCD04 contains two separate channels and supports 1.1V VDD supply. Designed for use on DDR5 RDIMMs, the chip can help the upcoming server systems address performance bottlenecks and provide critical support for demanding workloads” such as artificial intelligence, machine learning, and high-performance computing.):

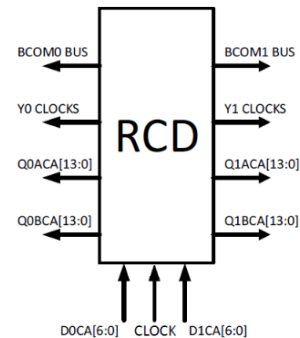
U.S. Patent No. 10,025,731 Domestic Industry Claim Chart (Samsung)

Claim 1

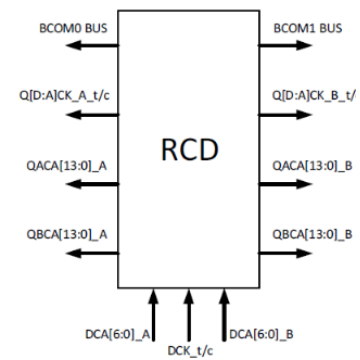
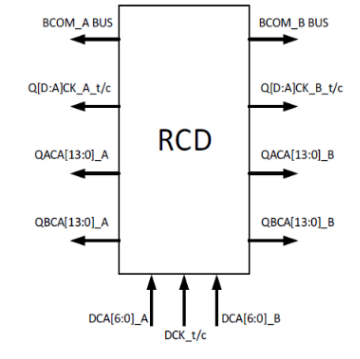
Evidence of Use

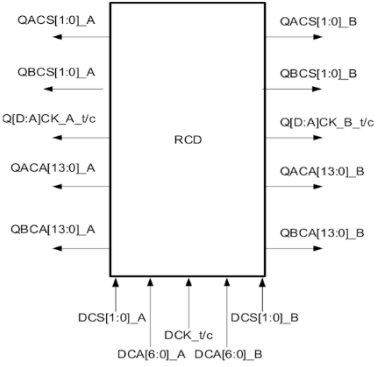


Ex. CC-52 (Gen 1 RCD Renesas 5RCD0148) at 1 (Gen 1 RCD by Renesas, another supplier to Samsung) (“The 5RCD0148HC2 (RCD) is a registering clock driver used on DDR5 RDIMMs and LRDIMMs. It supports DDR5 server speeds up to 4800 MT/s. Its primary function is to buffer the Command Address (CA) bus, chip selects, and clock between the host controller and the DRAMs. It also creates a BCOM bus to control the data buffers for LRDIMMs. The 5RCD0148HC2 contains two separate channels with some common logic such as clocking, but otherwise operate independently of each other. Each channel has a 7-bit double data rate CA bus input, a single parity input, two chip-select inputs, produces two copies of 14-bit single data rate CA bus outputs and two copies of the chip-select outputs. The RCD has a common clock input and PLL but produces separate clock outputs to the DRAM channels.”):



U.S. Patent No. 10,025,731 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	Ex. CC-51 (Gen 2 RCD Renesas RG5R256) at 1 (Gen 2 RCD):  Ex. CC-99 (Gen 3 RCD Renesas RG5R364) at 1 (Gen 3 RCD):  Ex. CC-100 (Gen 4 RCD Renesas RRG5004) at 1 (Gen 4 RCD by Renesas):

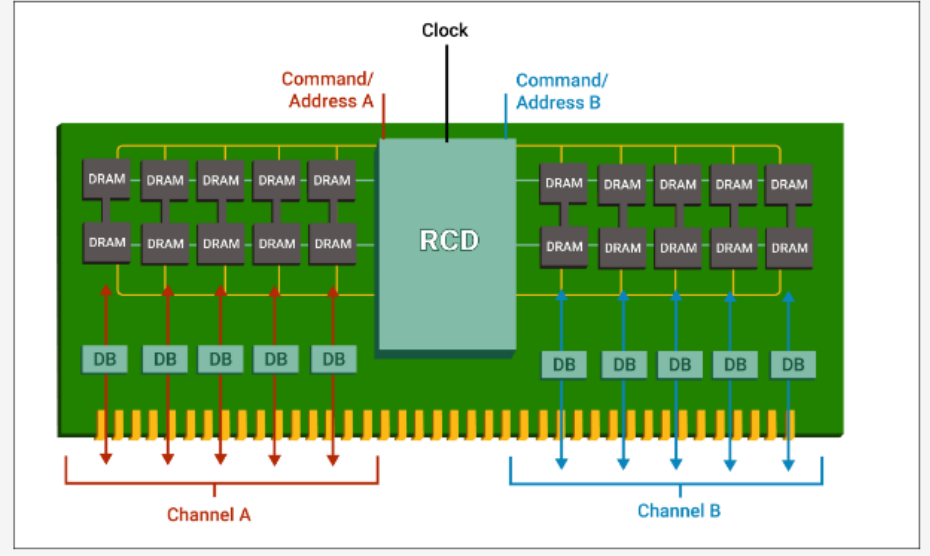
Claim 1	Evidence of Use
	Block Diagram  A DDR5 SDRAM on a memory module receives data from the memory controller (write operation) or outputs data to the memory controller in response to the control and address signals it receives, which are transmitted either directly or through an RCD to the address/control signal input circuitry of the SDRAM. <i>See</i> Ex. CC-46 (DDR5 Command Truth Table) (showing that commands for Activate, Read, and Write are composed of C/A[0:13] and CS signals):

U.S. Patent No. 10,025,731 Domestic Industry Claim Chart (Samsung)

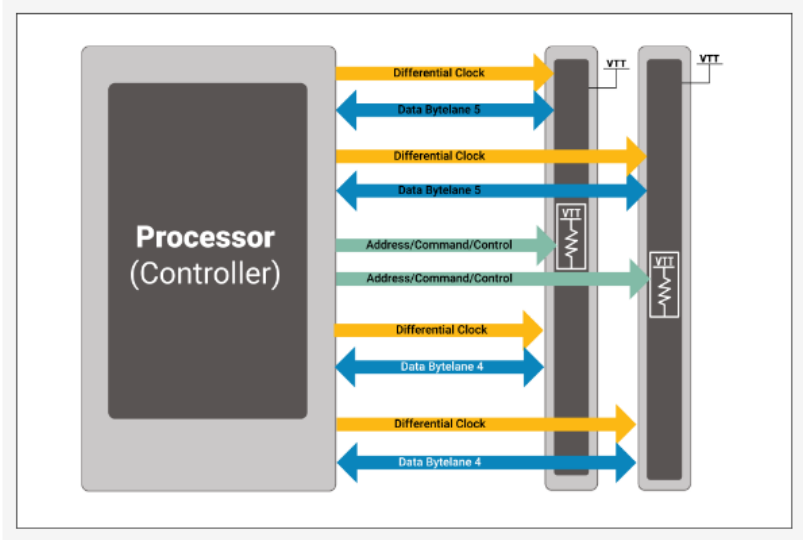
Claim 1	Evidence of Use																	
Function	Abbrevia- tion	CS_n	CA Pins													NOTES		
			CA0	CA1	CA2	CA3	CA4	CA5	CA6	CA7	CA8	CA9	CA10	CA11	CA12		CA13	
Activate	ACT	L	L	L	R0	R1	R2	R3	BA0	BA1	BG0	BG1	BG2	CID0	CID1	CID2/ DDPIO	11,17,20,21, 23,24	
		H	R4	R5	R6	R7	R8	R9	R10	R11	R12	R13	R14	R15	R16	CID3/ R17		
RFU	RFU	L	H	L	L	L	L	V	V	V	V	V	V	V	V	V		
		H	V	V	V	V	V	V	V	V	V	V	V	V	V	V		
RFU	RFU	L	H	L	L	L	H	V	V	V	V	V	V	V	V	V		
		H	V	V	V	V	V	V	V	V	V	V	V	V	V	V		
Write Pattern	WRP	L	H	L	L	H	L	H	BA0	BA1	BG0	BG1	BG2	CID0	CID1	CID2/ DDPIO	11,15,18,19, 20,21,23,24	
		H	V	C3	C4	C5	C6	C7	C8	C9	C10	V	H	H	V	CID3		
Write Pattern w/ Auto Precharge	WRPA	L	H	L	L	H	L	H	BA0	BA1	BG0	BG1	BG2	CID0	CID1	CID2/ DDPIO	11,15,18,19, 20,21,23,24	
		H	V	C3	C4	C5	C6	C7	C8	C9	C10	V	AP=L	H	V	CID3		
RFU	RFU	L	H	L	L	H	H	V	V	V	V	V	V	V	V	V		
		H	V	V	V	V	V	V	V	V	V	V	V	V	V	V		
Mode Register Write	MRW	L	H	L	H	L	L	MRA0	MRA1	MRA2	MRA3	MRA4	MRA5	MRA6	MRA7	V/ DDPIO	8,11,13,20, 22,23,24	
		H	OP0	OP1	OP2	OP3	OP4	OP5	OP6	OP7	V	V	CW	V	V	V		
Mode Register Read	MRR	L	H	L	H	L	H	MRA0	MRA1	MRA2	MRA3	MRA4	MRA5	MRA6	MRA7	V/ DDPIO	8,13,20,22, 23,24,25,26	
		H	L	L	V	V	V	V	V	V	V	V	CW	V	V	V		
Write	WR	L	H	L	H	H	L	BL*=L	BA0	BA1	BG0	BG1	BG2	CID0	CID1	CID2/ DDPIO	8,12,15,19, 20,21,23,24	
		H	V	C3	C4	C5	C6	C7	C8	C9	C10	V	H	WRL Partial=L	V	CID3		
Write w/Auto Precharge	WRA	L	H	L	H	H	L	BL*=L	BA0	BA1	BG0	BG1	BG2	CID0	CID1	CID2/ DDPIO	8,12,15,19, 20,21,23,24	
		H	V	C3	C4	C5	C6	C7	C8	C9	C10	V	AP=L	WRL Partial=L	V	CID3		
Read	RD	L	H	L	H	H	H	BL*=L	BA0	BA1	BG0	BG1	BG2	CID0	CID1	CID2/ DDPIO	8,15,19,20, 21,23,24	
		H	C2	C3	C4	C5	C6	C7	C8	C9	C10	V	H	V	V	CID3		
Read w/Auto Precharge	RDA	L	H	L	H	H	H	BL*=L	BA0	BA1	BG0	BG1	BG2	CID0	CID1	CID2/ DDPIO	8,15,19,20, 21,23,24	
		H	C2	C3	C4	C5	C6	C7	C8	C9	C10	V	AP=L	V	V	CID3		

See also Ex. CC-47 (DDR Memory & Challenges in PCB Design) at 8:

U.S. Patent No. 10,025,731 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	 The diagram illustrates a dual-channel memory architecture. A central RCD (Row Control Driver) block is connected to two channels, Channel A and Channel B. Channel A is on the left, and Channel B is on the right. Each channel contains two rows of DRAM (Dynamic Random Access Memory) and two rows of DB (Data Buffer) blocks. The RCD block is connected to the DRAM blocks via Command/Address lines (A and B) and a Clock line. The DB blocks are connected to the DRAM blocks via Data lines (A and B). The diagram shows the flow of data and control signals between the channels and the RCD block.

U.S. Patent No. 10,025,731 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	Bus topologies—On-board two-UDIMM  Ex. CC-47 (DDR Memory & Challenges in PCB Design) at 22–23. This claim element is literally present, as described herein, but is also satisfied under the doctrine of equivalents, and the Samsung Practicing '731 Products are manufactured in a manner that is not substantially different from the requirements of this limitation because they achieve substantially or exactly the same function, in substantially or exactly the same way as set forth in the claim element (as described herein), and achieve substantially or exactly the same result.

U.S. Patent No. 10,025,731 Domestic Industry Claim Chart (Samsung)

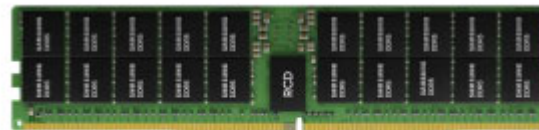
[1.b] [i] a printed circuit board comprising [ii] at least one connector [iii] configured to be operatively coupled to the memory bus, [iv] the at least one connector including a plurality of edge connections distributed along one or more edges of the printed circuit board;

[i] Samsung's DDR5 RDIMMs and MRDIMMs each further comprise *a printed circuit board* (i.e., the green board on which the DDR5 memory devices, the RCD, the SPD Hub, the thermal sensors, and other module components are mounted):

See, e.g., Ex. CC-94 (Samsung Module Handling Guide) at 2:

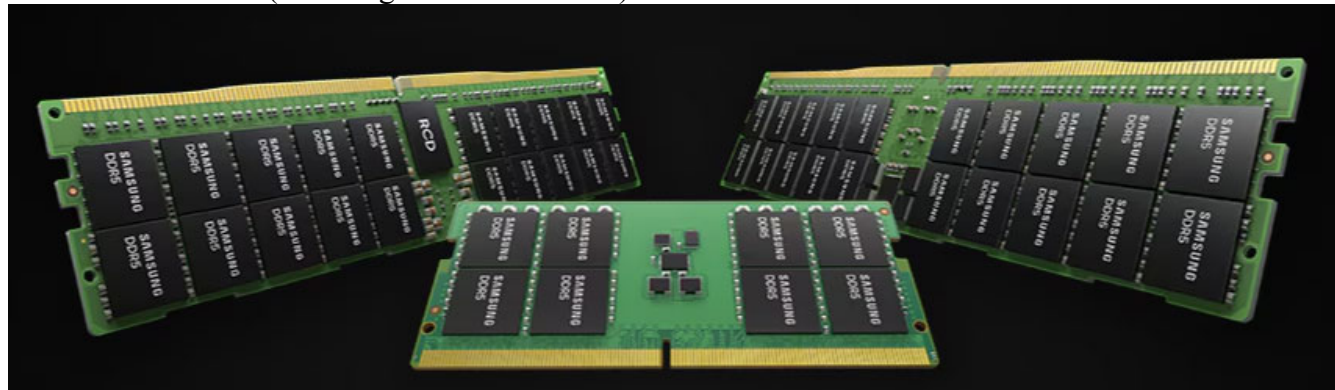


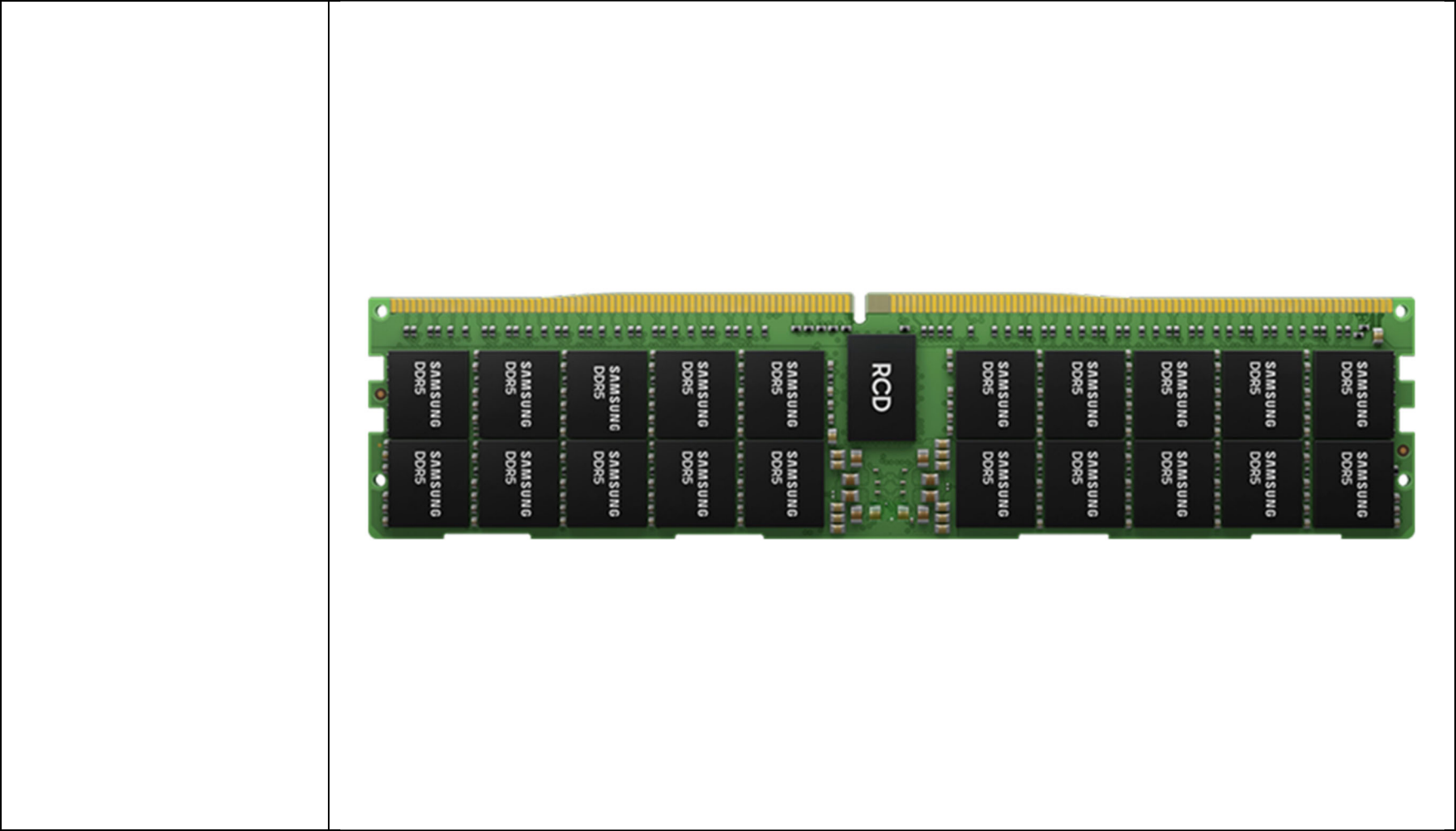
Multiplexed Rank DIMM



Registered DIMM

See also Ex. CC-95 (Samsung DRAM Modules) at 1–2:

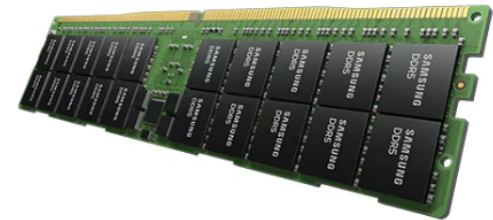




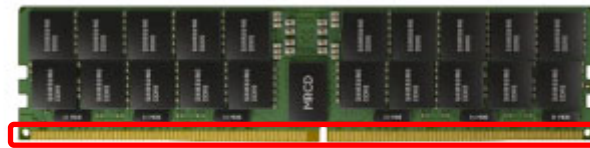
RDIMM

Registered DIMM

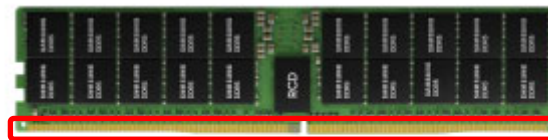
Include a register for enhancing the clock,
command and control signals
Error correction is available by adding 8-bit parity signals
Supports x4 / x8 Organization / up to 2 ranks per DIMM and
3DPC configuration
Application : Server



[ii] The printed circuit board comprises *at least one connector*. For example, as shown in the images above and as annotated below, each Samsung memory module's printed circuit board has a connector composed of two rows of gold fingers (one on the front side of PCB and the other on the backside of PCB) at the bottom edge of the printed circuit board:



Multiplexed Rank DIMM



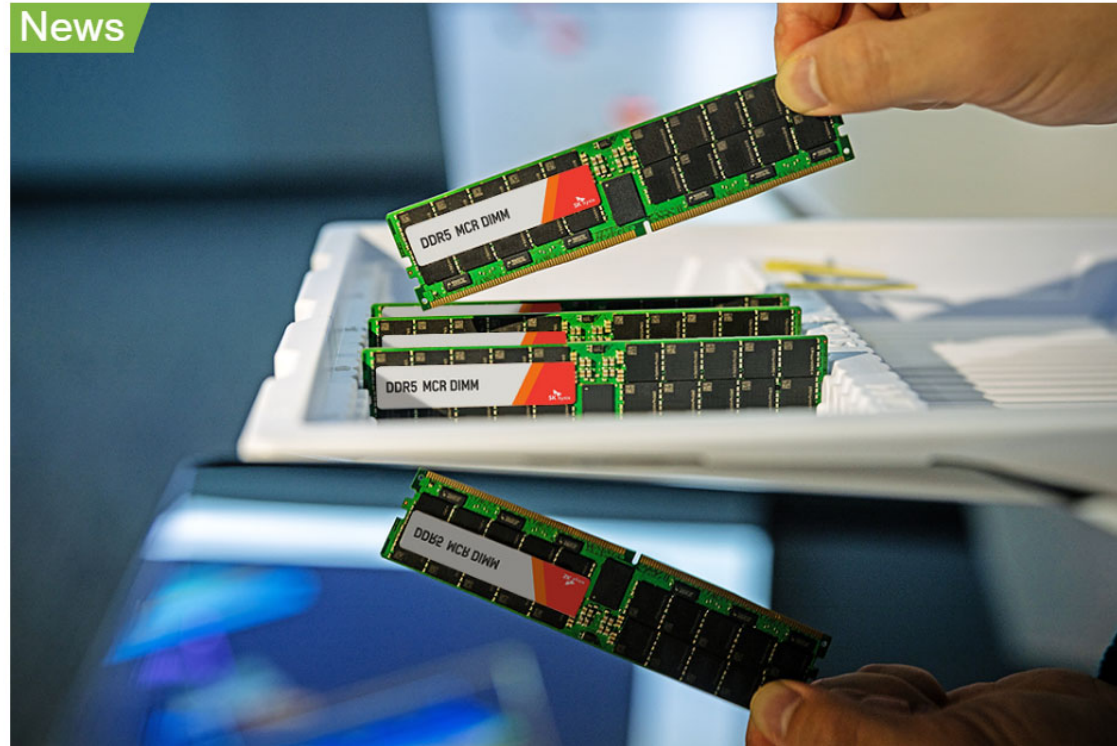
Registered DIMM

Ex. CC-94 (Samsung Module Handling Guide) at 2 (annotated).

[iii] The at least one connector [the edge connector] is *configured to be operatively coupled to the*

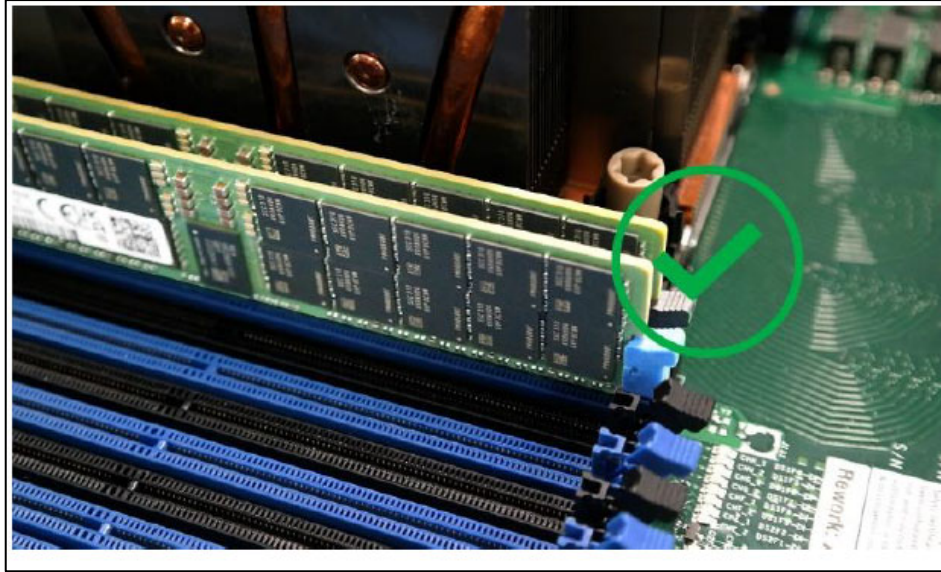
U.S. Patent No. 10,025,731 Domestic Industry Claim Chart (Samsung)

memory bus. As would be understood by artisans in the field, the edge connector is to be inserted into a memory slot for coupling to the two memory channels (*i.e.*, the communication channels between the CPU and the SDRAMs on the memory module) connected to the system memory controller, as illustrated below:



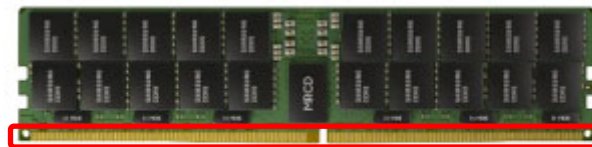
Ex. CC-50 (TrendForce News MRDIMM-MCRDIMM) at 1.

U.S. Patent No. 10,025,731 Domestic Industry Claim Chart (Samsung)



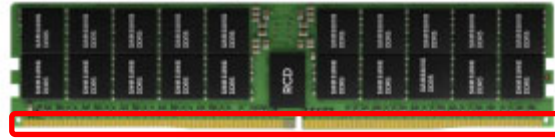
Ex. CC-94 (Samsung Module Handling Guide) at 9.

[iv] The at least one connector includes *a plurality of edge connections distributed along one or more edges of the printed circuit board* (e.g., as edge pins or gold fingers placed along the bottom edges in the DIMM images below).



Multiplexed Rank DIMM

U.S. Patent No. 10,025,731 Domestic Industry Claim Chart (Samsung)



Registered DIMM

Ex. CC-94 (Samsung Module Handling Guide) at 2 (annotated).

For example, the Samsung DDR5 RDIMM Datasheet at page 6 shows 288 pins distributed along the bottom edge of the printed circuit board:

U.S. Patent No. 10,025,731 Domestic Industry Claim Chart (Samsung)

4.0 REGISTERED DIMM PIN INFORMATION

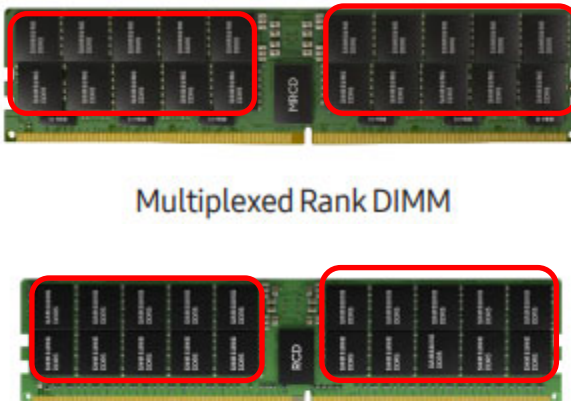
[Table 4] PIN Configuration

Pin	Front Side	Pin	Back Side	Pin	Front Side	Pin	Back Side	Pin	Front Side	Pin	Back Side
1	VIN_BULK	145	VIN_BULK	50	VSS	194	DQ31_A	96	CB0_B	240	VSS
2	RFU	146	VIN_BULK	51	CB0_A	195	VSS	97	VSS	241	CB2_B
3	VIN_MGMT	147	PWR_GOOD/FAIL_n	52	VSS	196	CB2_A	98	CB1_B	242	VSS
4	SCL	148	SAA	53	CB1_A	197	VSS	99	VSS	243	CB3_B
5	SDA	149	RFU	54	VSS	198	CB3_A	100	DQ0_B	244	VSS
6	VSS	150	RFU	55	DQ04_A_I	199	VSS	101	VSS	245	DQ2_B
7	DQ0_A	151	VSS	56	DQ04_A_c	200	DQ09_A_cTDQ09_A_c	102	DQ1_B	246	VSS
8	VSS	152	DQ2_A	57	VSS	201	DQ09_A_cTDQ09_A_c	103	VSS	247	DQ3_B
9	DQ1_A	153	VSS	58	CB4_A	202	VSS	104	DQ02_B_I	248	VSS
10	VSS	154	DQ3_A	59	VSS	203	CB0_A	105	DQ00_B_c	249	DQ05_B_cTDQ05_B_c
11	DQ00_A_I	155	VSS	60	CB5_A	204	VSS	106	VSS	250	DQ05_B_cTDQ05_B_c
12	DQ00_A_c	156	DQ05_A_cTDQ05_A_c	61	VSS	205	CB7_A	107	DQ4_B	251	VSS
13	VSS	157	DQ05_A_cTDQ05_A_c	62	ALERT_n	206	VSS	108	VSS	252	DQ6_B
14	DQ4_A	158	VSS	63	VSS	207	RESET_n	109	DQ5_B	253	VSS
15	VSS	159	DQ6_A	64	CB0_A_n	208	VSS	110	VSS	254	DQ7_B
16	DQ5_A	160	VSS	65	VSS	209	CB1_A_n	111	DQ6_B	255	VSS
17	VSS	161	DQ7_A	66	CA0_A	210	VSS	112	VSS	256	DQ10_B
18	DQ6_A	162	VSS	67	VSS	211	CA1_A	113	DQ8_B	257	VSS
19	VSS	163	DQ10_A	68	CA2_A	212	VSS	114	VSS	258	DQ11_B
20	DQ6_A	164	VSS	69	VSS	213	CA3_A	115	DQ01_B_I	259	VSS
21	VSS	165	DQ11_A	70	CA4_A	214	VSS	116	DQ01_B_c	260	DQ06_B_cTDQ06_B_c
22	DQ01_A_I	166	VSS	71	VSS	215	CA5_A	117	VSS	261	DQ06_B_cTDQ06_B_c
23	DQ01_A_c	167	DQ06_A_cTDQ06_A_c	72	CA6_A	216	VSS	118	DQ12_B	262	VSS
24	VSS	168	DQ06_A_cTDQ06_A_c	73	VSS	217	CK_I	119	VSS	263	DQ14_B
25	DQ12_A	169	VSS	74	PAR_A	218	CK_c	120	DQ13_B	264	VSS
26	VSS	170	DQ14_A	75	VSS	219	VSS	121	VSS	265	DQ15_B
27	DQ13_A	171	VSS	KEY				122	DQ16_B	266	VSS
28	VSS	172	DQ15_A					123	VSS	267	DQ16_B
29	DQ16_A	173	VSS					124	DQ17_B	268	VSS
30	VSS	174	DQ16_A	76	CA0_B	220	RFU	125	VSS	269	DQ19_B
31	DQ17_A	175	VSS	77	VSS	221	CA1_B	126	DQ02_B_I	270	VSS
32	VSS	176	DQ16_A	78	CA2_B	222	VSS	127	DQ02_B_c	271	DQ07_B_cTDQ07_B_c
33	DQ02_A_I	177	VSS	79	VSS	223	CA3_B	128	VSS	272	DQ07_B_cTDQ07_B_c
34	DQ02_A_c	178	DQ07_A_cTDQ07_A_c	80	CA4_B	224	VSS	129	DQ20_B	273	VSS
35	VSS	179	DQ07_A_cTDQ07_A_c	81	VSS	225	CA5_B	130	VSS	274	DQ22_B
36	DQ20_A	180	VSS	82	CA6_B	226	VSS	131	DQ21_B	275	VSS
37	VSS	181	DQ22_A	83	VSS	227	PAR_B	132	VSS	276	DQ23_B
38	DQ21_A	182	VSS	84	CB0_B_n	228	VSS	133	DQ24_B	277	VSS
39	VSS	183	DQ23_A	85	VSS	229	CB1_B_n	134	VSS	278	DQ26_B
40	DQ24_A	184	VSS	86	LB0_RSP_A_n	230	VSS	135	DQ25_B	279	VSS
41	VSS	185	DQ26_A	87	LB8_RSP_B_n	231	RFU	136	VSS	280	DQ27_B
42	DQ25_A	186	VSS	88	VSS	232	RFU	137	DQ03_B_I	281	VSS
43	VSS	187	DQ27_A	89	CB4_B	233	VSS	138	DQ03_B_c	282	DQ08_B_cTDQ08_B_c
44	DQ03_A_I	188	VSS	90	VSS	234	CB6_B	139	VSS	283	DQ08_B_cTDQ08_B_c
45	DQ03_A_c	189	DQ08_A_cTDQ08_A_c	91	CB5_B	235	VSS	140	DQ28_B	284	VSS
46	VSS	190	DQ08_A_cTDQ08_A_c	92	VSS	236	CB7_B	141	VSS	285	DQ30_B
47	DQ28_A	191	VSS	93	DQ09_B_cTDQ09_B_c	237	VSS	142	DQ29_B	286	VSS
48	VSS	192	DQ30_A	94	DQ09_B_cTDQ09_B_c	238	DQ04_B_c	143	VSS	287	DQ31_B
49	DQ29_A	193	VSS	95	VSS	239	DQ04_B_I	144	RFU	288	VSS

Ex. CC-89 (Samsung DDR5 RDIMM Datasheet) at 6.

U.S. Patent No. 10,025,731 Domestic Industry Claim Chart (Samsung)

	This claim element is literally present, as described herein, but is also satisfied under the doctrine of equivalents, and the Samsung Practicing '731 Products are manufactured in a manner that is not substantially different from the requirements of this limitation because they achieve substantially or exactly the same function, in substantially or exactly the same way as set forth in the claim element (as described herein), and achieve substantially or exactly the same result.
--	---

Claim 1	Evidence of Use
[1.c] [i] a plurality of memory devices on the printed circuit board, [ii] the plurality of memory devices being arranged in multiple ranks, [iii] each rank comprising an independent set of memory devices that can be accessed by the system memory controller to access a full bit-width of the memory bus;	[i] Samsung's DDR5 RDIMMs and MRDIMMs each include <i>a plurality of memory devices</i> (e.g., memory cores of Samsung DDR5 SDRAMs) <i>on the printed circuit board</i>, which are located within the packaged SDRAMs annotated below.  Multiplexed Rank DIMM Registered DIMM Ex. CC-94 (Samsung Module Handling Guide) at 2 (annotated). As an example, each SDRAM may include a “memory device,” a circuit of the at least one circuit, and

U.S. Patent No. 10,025,731 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	potentially a portion of the control circuit. DDR5 SDRAM [ii] The plurality of memory devices (e.g., memory cores of Samsung DDR5 SDRAMs) <i>are arranged in multiple ranks</i>. As stated on Samsung’s web site, Samsung supports “up to 2 ranks per DIMM” for its DDR5 SDRAMs. Ex. CC-95 (Samsung DRAM Modules) at 2. However, DDR5 modules have also adopted a dual channel configuration, each channel having its own independent set of control/address signals and data/strobe signals. Cf., Ex. CC-47 (DDR Memory & Challenges in PCB Design) at 9 (“In DDR5, each DIMM will have two 40-bit channels (32 data bits and 8 ECC bits). While the data width is the same (64-bits total), having two smaller independent channels improves DDR5 memory access efficiency. The benefit of this is higher MT/s is amplified by greater efficiency.”); Ex. CC-98

U.S. Patent No. 10,025,731 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	(Montage M88DR5RCD04) at 1 (the RCD “contains two separate channels”); Ex. CC-97 (Montage M88DR5RCD03) (same); Ex. CC-87 (Montage M88DR5RCD02) (same); Ex. CC-48 (Montage M88DR5RCD01) at 1 (“a two-channel RCD designed for driving addresses and controlling nets between the host controller and the DDR5 SDRAMs”); Ex. CC-96 (Rambus DDR5 RCD) at 1 (emphasis added) (“In DDR5 DIMM architectures, the module is organized into two independent 40-bit subchannels (32 bits data, 8 bits ECC). <u>Each subchannel [for a RDIMM] is organized as either 10 devices in a x4 configuration with 2 ranks or 5 devices in a x8 configuration with 2 ranks</u>. The RCD distributes C/A signals and clocks for each subchannel, rank and nibble.”); Ex. CC-100 (Gen 4 RCD Renesas RRG5004) at 1 (“Dual-Channel Design: Operates independently with shared common logic such as clocking, enhancing performance and reliability.”); Ex. CC-99 (Gen 3 RCD Renesas RG5R364) at 1 (The RCD “contains two separate channels with some common logic such as clocking, but otherwise operate independently of each other.”); Ex. CC-51 (Gen 2 RCD Renesas RG5R256) (same); Ex. CC-52 (Gen 1 RCD Renesas 5RCD0148) (same); Ex. CC-101 (Kingston Technology DDR5 Overview) at 1 (emphasis added) (“DDR5 splits the memory module into two independent 32-bit addressable subchannels to increase efficiency and lower the latencies of data accesses for the memory controller. The data width of the DDR5 module is still 64-bit, however breaking it down into two 32-bit addressable channels increases overall performance. For server class memory (RDIMMs), 8-bits are added to each subchannel for ECC support for a total of 40-bits per subchannel, or 80-bits per Rank. Dual Rank modules feature four 32-bit subchannels.”). Because SDRAMs in channel A are addressed independently from SDRAMs in channel B, the SDRAMs in what Samsung calls a “rank” may also be viewed as consisting of two independent groups, each having a number of SDRAMs, each SDRAM having memory cores. For example, on what Samsung characterizes as a 1R memory module, there are two independent groups: (1) Rank0, channel A and (2) Rank0, channel B. As another example, on what Samsung characterizes as a 2R

U.S. Patent No. 10,025,731 Domestic Industry Claim Chart (Samsung)

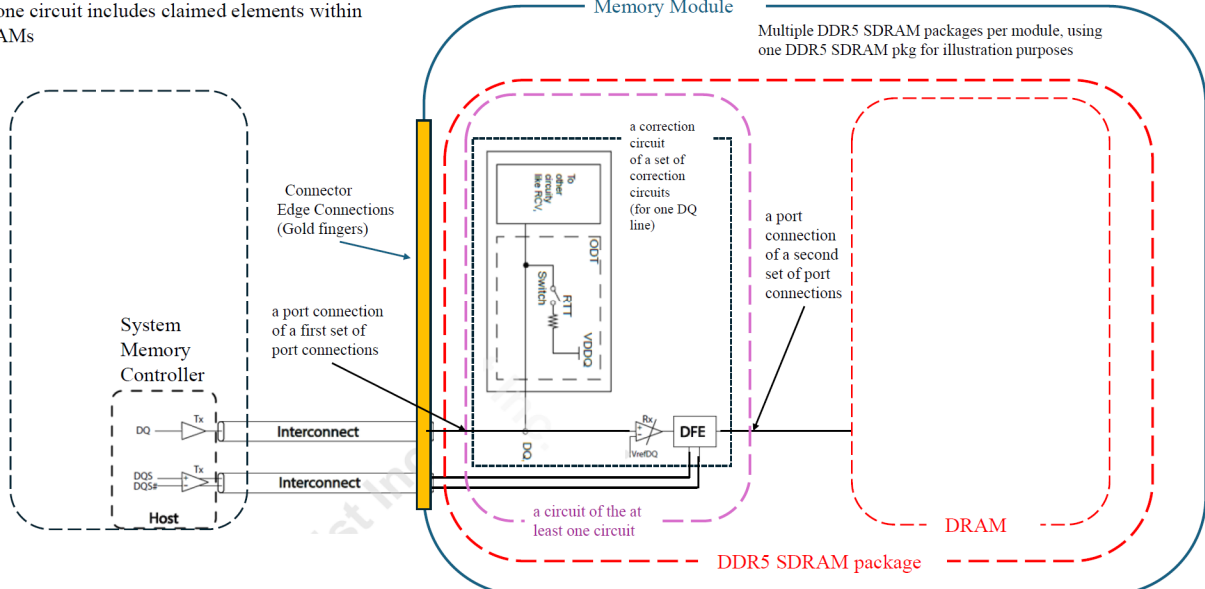
Claim 1	Evidence of Use								
	memory module, there are four independent groups: (1) Rank0, channel A, (2) Rank0, channel B, (3) Rank1, channel A; and (4) Rank1, channel B. Each of these groups may also be viewed as meeting the traditional definition of a “rank.” Ex. CC-102 (IPR2023-00203, EX1033) at 318 (“Each rank is a set of DRAM devices that operate in unison[.]”). [iii] <i>Each rank compris[es] an independent set of memory devices that can be accessed by the system memory controller to access a full bit-width of the memory bus (e.g., x32-x40 per channel or x64-x80 for two channels).</i> Where the communication channel width is considered to be x64—x80, then a group of memory devices corresponding to what Samsung calls a rank is accessed by the system memory controller to access that full width of the memory bus, albeit through two independent commands. On the other hand, if each communication channel is considered to be x32—x40 (<i>i.e.</i>, the width of channel A and channel B), then a group of devices corresponding to a channel as described above is accessed by the system memory controller to access the x32—x40 width of the memory bus. <i>See also</i> Ex. CC-89 (Samsung DDR5 RDIMM Datasheet) at 11 (annotated): <table><tr><th>Symbol</th><th>Type</th><th>I/O Levels</th><th>Function</th></tr><tr><td>CS[1:0]_A CS[1:0]_B</td><td>Input</td><td>VDD</td><td>Chip Select : All commands are masked when CS_n is registered HIGH. CS_n provides for external Rank selection on systems with multiple Ranks. CS_n is considered part of the command code. CS_n is also used to enter and exit the parts from power down mode and self refresh mode. While not in self refresh mode the CS_n input buffer operates with the same ODT and VREF parameters as configured by the CA_ODT strap setting or mode register. When in self refresh the CS_n is a CMOS rail to rail signal with DC high and low at 80% and 20% of VDD.</td></tr></table> This claim element is literally present, as described herein, but is also satisfied under the doctrine of equivalents, and the Samsung Practicing ’731 Products are manufactured in a manner that is not substantially different from the requirements of this limitation because they achieve substantially or	Symbol	Type	I/O Levels	Function	CS[1:0]_A CS[1:0]_B	Input	VDD	Chip Select : All commands are masked when CS_n is registered HIGH. CS_n provides for external Rank selection on systems with multiple Ranks. CS_n is considered part of the command code. CS_n is also used to enter and exit the parts from power down mode and self refresh mode. While not in self refresh mode the CS_n input buffer operates with the same ODT and VREF parameters as configured by the CA_ODT strap setting or mode register. When in self refresh the CS_n is a CMOS rail to rail signal with DC high and low at 80% and 20% of VDD.
Symbol	Type	I/O Levels	Function						
CS[1:0]_A CS[1:0]_B	Input	VDD	Chip Select : All commands are masked when CS_n is registered HIGH. CS_n provides for external Rank selection on systems with multiple Ranks. CS_n is considered part of the command code. CS_n is also used to enter and exit the parts from power down mode and self refresh mode. While not in self refresh mode the CS_n input buffer operates with the same ODT and VREF parameters as configured by the CA_ODT strap setting or mode register. When in self refresh the CS_n is a CMOS rail to rail signal with DC high and low at 80% and 20% of VDD.						

U.S. Patent No. 10,025,731 Domestic Industry Claim Chart (Samsung)

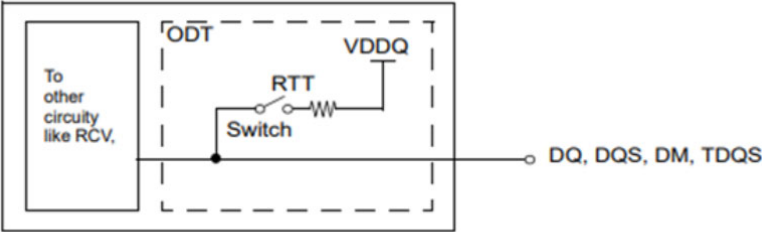
Claim 1	Evidence of Use
	exactly the same function, in substantially or exactly the same way as set forth in the claim element (as described herein), and achieve substantially or exactly the same result.

Claim 1	Evidence of Use
[1.d] [i] at least one circuit coupled between the at least one connector and the plurality of memory devices, [ii] the at least one circuit comprising a first set of port connections coupled to respective ones of the plurality of edge connections and a second set of port connections coupled to the plurality of memory devices,	[i] Samsung DDR5 RDIMMs and MRDIMMs each include at least one circuit (e.g., a collection of the DDR5-specific DFE and ODT/RTT circuits on each of the DDR5 SDRAMs) coupled between the at least one connector (e.g., the edge connectors) and the plurality of memory devices (the memory cores of DDR5 SDRAMs devices), as illustrated below. <i>See supra</i> limitations [1.b]-[1.c]. For example, as shown in the Samsung DDR5 RDIMM Datasheet, there is a DFE circuit associated with each DQ line on a DRAM (DDR5 SDRAM), coupled to the host through the edge connector. Figure 17. Example of DDR5 Memory Interconnect Ex. CC-89 (Samsung DDR5 RDIMM Datasheet) at 49. An illustration is provided below for a circuit associated with one pin within the at least one circuit:

U.S. Patent No. 10,025,731 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	The at least one circuit includes claimed elements within DDR5 SDRAMs  In addition, on information and belief, each SDRAM is expected to include on-die termination (“ODT”) circuitry, since “ODT is a crucial component for high-speed DDR5 memory systems since it aids in enhancing signal quality, decreasing signal ringing, and eventually allowing for higher data transfer speeds with less signal deterioration.” Ex. CC-10 (Intellect Partners DDR5’s Secret Weapon_ODT) at 2; <i>see also</i> Ex. CC-103 <i>Netlist, Inc. v. Samsung Electronics Co., Ltd. et al.</i>, No. 2:25-cv-00557-JRG, Dkt. No. 14-6 (E.D. Tex. July 8, 2025) (“FAC Ex. 6”).

U.S. Patent No. 10,025,731 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	 [Source: DDR5 Standard [JEDEC JESD79-5B_v1.20] Page 346 of 502] Reproduced from Ex. CC-10 (Intellect Partners DDR5’s Secret Weapon_ODT) which in turn cites from JEDEC JESD79-5B_v1.20. On information and belief, Samsung’s DDR5 SDRAMs comply with the JEDEC requirements. <i>E.g.</i>, Ex. CC-104 (Samsung Optimized DDR5 DIMM Solutions) at (“Samsung manufactures products that abide by JEDEC standards.”). In addition, DDR5 introduced dynamic ODT, by which the termination resistor “can be changed or turned on or off” depending on “the settings of the memory controller and the precise data transfer requirements at any given time,” all without mode register set (MRS) instructions. Ex. CC-10 (Intellect Partners DD5’s Secret Weapon_ODT) at 3. Dynamic ODT permits DDR5 memory system to switch between high and low termination impedance. <i>Id.</i> Additionally, the dynamic management of the termination impedance offered by dynamic ODT “optimizes power usage by changing the termination parameters in accordance with the demands of the data transfer,” therefore reducing the amount of unnecessary power dissipation and improving the power efficiency of the memory system. <i>Id.</i> at 5. To the extent the whole or part of the DFE and/or ODT/RTT circuitry resides outside the SDRAM chips (e.g., in multiplexing data buffers (MDBs) of the MRDIMM Products),¹ that location would still be coupled between the at least one connector (e.g., the edge connectors on the printed circuit board)

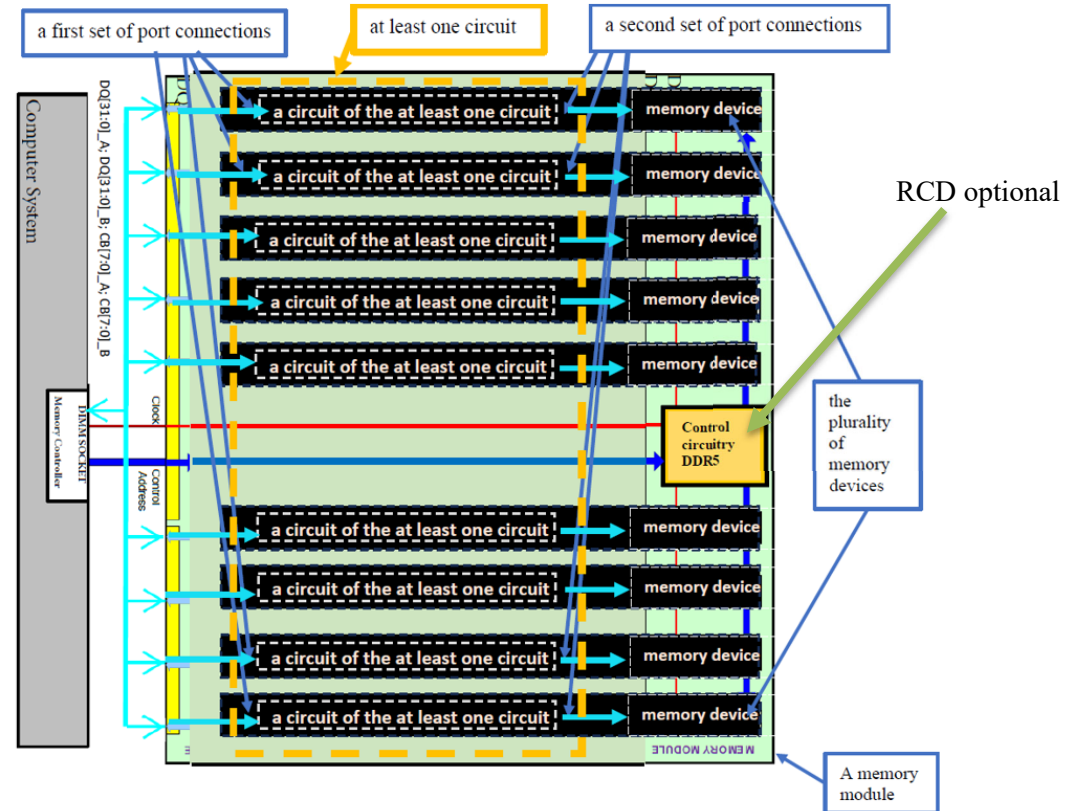
¹ Montage, a supplier of MDBs, reported: “We have introduced multiple innovative circuits and algorithms to address the signal integrity challenges of DDR5’s high-speed parallel bus. On the circuit level, we developed high-speed, low-noise transceivers, supplemented by multi-tap decision feedback equalization (DFE) to compensate for far-end crosstalk and inter-symbol interference.” Ex. CC-71 (Montage HK Proof) at 180.

U.S. Patent No. 10,025,731 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	and the plurality of memory devices (e.g., the memory cores of the DDR5 SDRAM chips). Specifically, if the DFE and/or ODT/RTT circuitry is in MDBs or some other circuit external to the SDRAM chips, that circuit would be coupled between the PCB's edge connectors and the SDRAM chips. In this configuration, the memory cores of the DDR5 SDRAM chips <i>or</i> the SDRAM chip packages as a whole would be the claimed "memory device." [ii] The at least one circuit (e.g., the collection of the DDR5-specific DFE and ODT/RTT circuits on each of the DDR5 SDRAMs) includes <i>a first set of port connections</i> (e.g., data ports on the host side) <i>coupled to respective ones of the plurality of edge connections</i> (e.g., the respective pins for data and error correction) <i>and a second set of port connections</i> (e.g., data ports on the DRAM side) <i>coupled to the plurality of memory devices</i> (memory cores of DDR5 SDRAMs), as illustrated below based on our current understanding:

Claim 1

Evidence of Use



Note: the RCD control circuit is optional.

Of note, each DDR5 SDRAM device on a Samsung memory module includes a memory device comprising memory cores and associated circuits and a portion of the at least one circuit such as the DDR5-specific DFE and ODT/RTT circuits (*e.g.*, for DQ and C/A, CS, CLK signals), which is each a correction circuit. The collection of the DFE and ODT/RTT circuits associated with a DDR5 SDRAM forms a set of correction circuits (*e.g.*, each x4, x8 and x16 DDR5 SDRAM has respectively

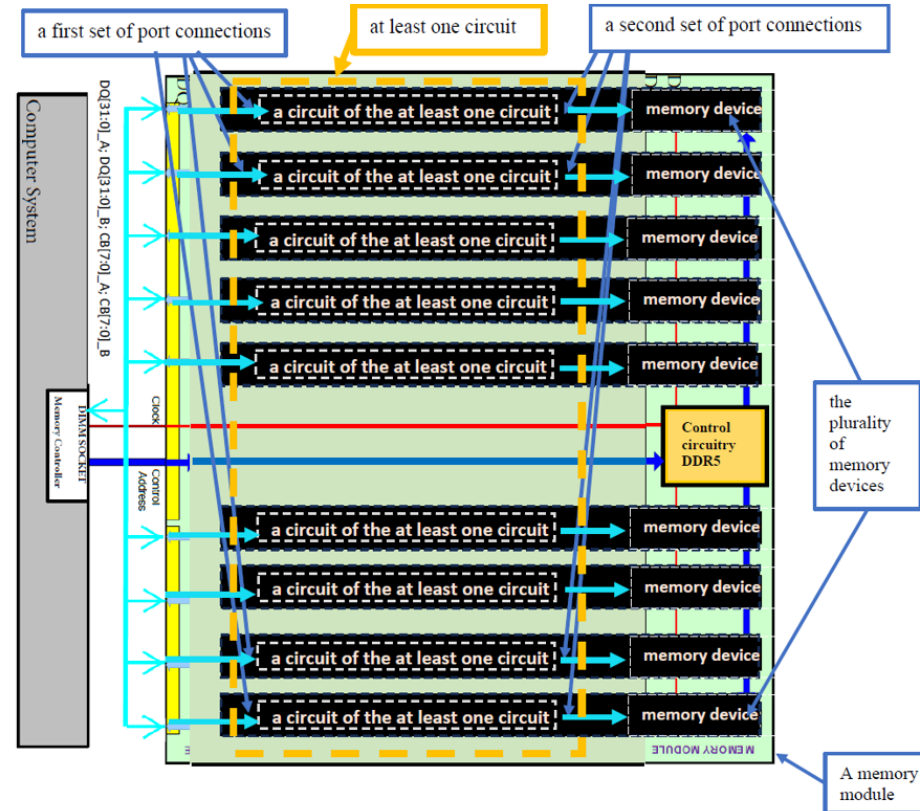
U.S. Patent No. 10,025,731 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	4, 8, and 16 DFE circuits associated with 16 DQ lines as well as those for C/A, CS and CLK signals). This claim element is literally present, as described herein, but is also satisfied under the doctrine of equivalents, and the Samsung Practicing '731 Products are manufactured in a manner that is not substantially different from the requirements of this limitation because they achieve substantially or exactly the same function, in substantially or exactly the same way as set forth in the claim element (as described herein), and achieve substantially or exactly the same result. For instance, whether the correction circuit is on the same silicon substrate as the memory core or in a separate package, it is on the same communication pathway between the host and the memory device (<i>e.g.</i>, the memory core and the associated control circuits), and performs the same correction function in the same way and achieves the same result.

Claim 1	Evidence of Use
[1.e] each circuit of the at least one circuit further comprising a set of correction circuits, each correction circuit of the set of correction circuits being configured to make corrections in one or more signals transmitted between a respective port connection of the first set of port connections and a corresponding port connection in the second set of port connections,	In Samsung's DDR5 RDIMMs and MRDIMMs, as explained above, the at least one circuit includes a collection of DFE and ODT/RTT circuitry on the SDRAMs. The at least one circuit includes multiple circuits, each maybe associated with one SDRAM, as illustrated below:

Claim 1

Evidence of Use

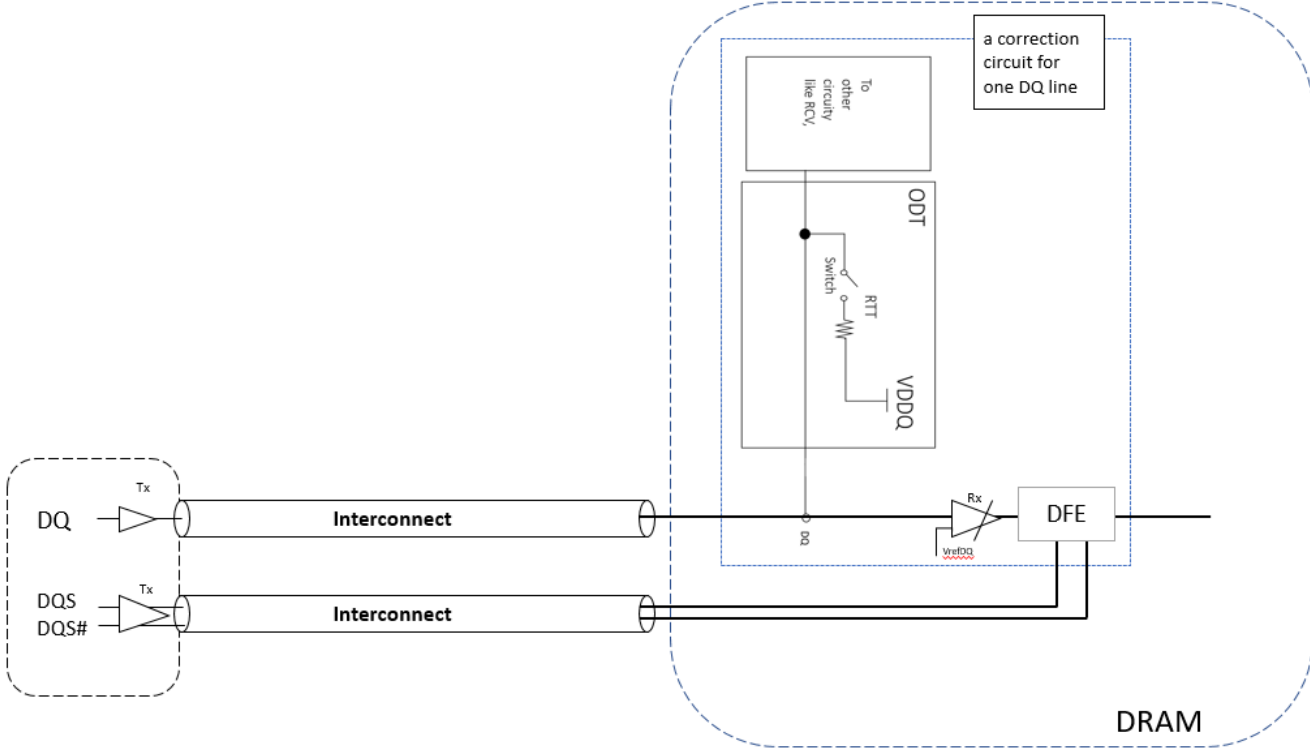


Note: the RCD control circuit is optional.

Each circuit of the at least one circuit (i.e., the DFE+ODT/RTT circuit associated with an SDRAM) ***further comprises a set of correction circuits*** (one correction circuit per DQ path and each memory device has at least 4 DQ paths). As noted in connection with element 1d, each DDR5 SDRAM device includes a memory device comprising memory cores (memory arrays) and associated circuits as well as DDR5-specific DFE and ODT/RTT circuits (e.g., for DQ and C/A, CS, CLK signals).

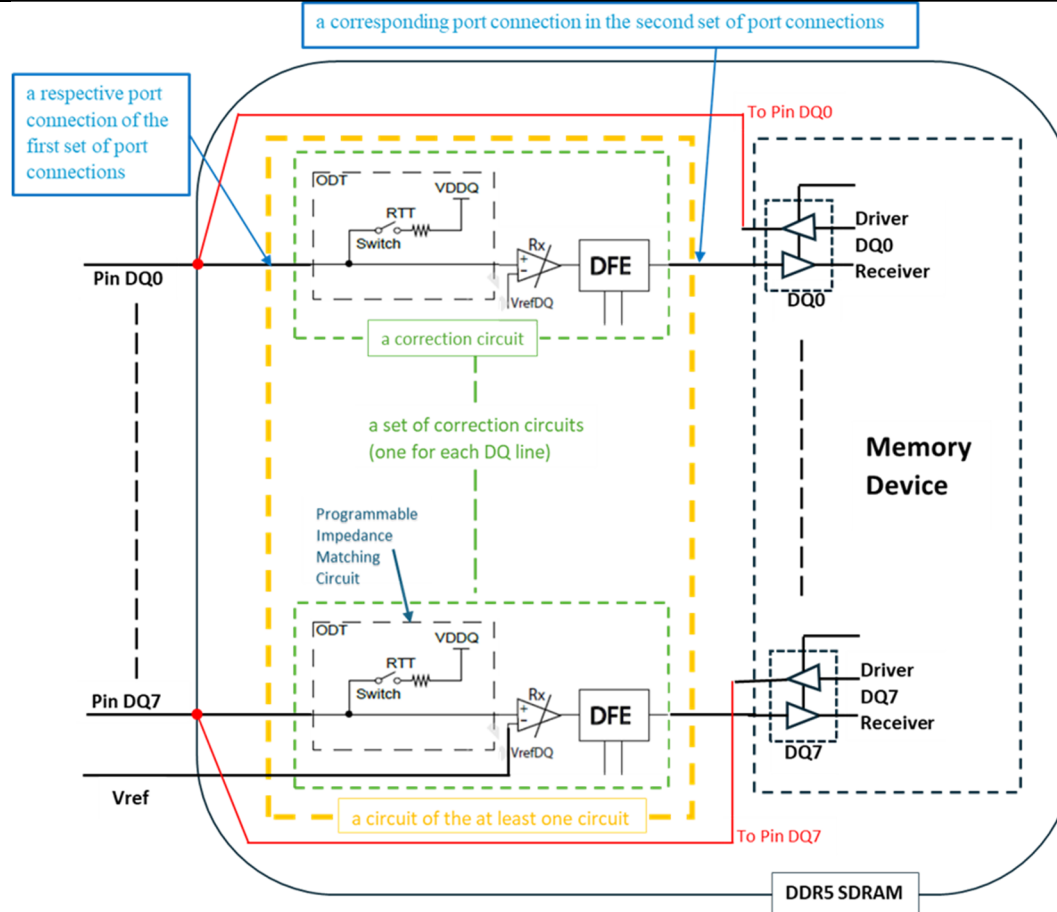
Claim 1	Evidence of Use
	Each DQ line has a correction circuit comprised of a DDR5-specific DFE circuit and ODT/RTT circuit. For example, as shown in the Samsung DDR5 RDIMM Datasheet, there is a DFE circuit associated with each DQ line on a DRAM (DDR5 SDRAM), coupled to the host through the edge connector, to ensure proper eye width. Figure 17. Example of DDR5 Memory Interconnect Ex. CC-89 (Samsung DDR5 RDIMM Datasheet) at 49. And as noted in limitation [1d], each DQ line also has an associated ODT circuit given the importance of the circuit in signal reflection and integrity. Because the memory devices on Samsung's DDR5 RDIMMs are each x4, x8 or x16 (see Ex. CC-95 (Samsung DRAM Modules)) and because there is a correction circuit for each DQ line, each SDRAM has 4, 8 or 16 correction circuits for the DQ lines. Thus, each circuit of the at least one circuit (which is a circuit associated with one SDRAM) further comprises a set of correction circuits (i.e., a set of 4, 8 or 16 correction circuits for the DQ lines). That is, the collection of the DFE and ODT/RTT circuits associated with a DDR5 SDRAM forms a set of correction circuits (e.g., each x4, x8 and x16 DDR5 SDRAM has respectively 4, 8, and 16 DFE circuits associated with 16 DQ lines). An illustration of a correction circuit is provided below:

U.S. Patent No. 10,025,731 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	 Further illustrated below is the set of correction circuits in each circuit of the at least one circuit for a x8 memory device, on information and belief:

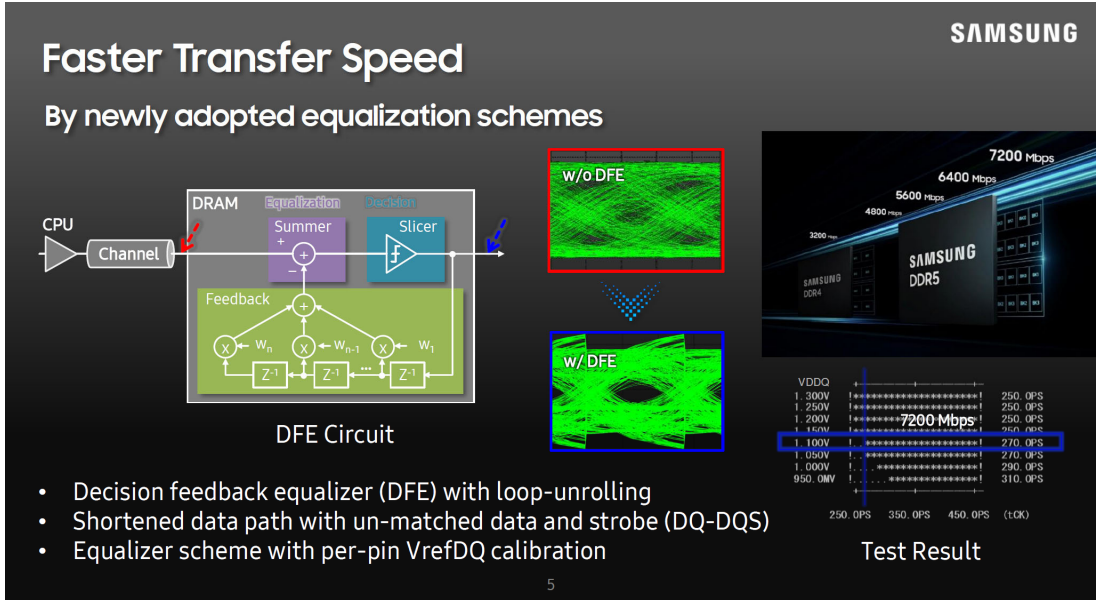
Claim 1

Evidence of Use



Each correction circuit of the set of correction circuits being configured to make corrections in one or more signals transmitted between a respective port connection of the first set of port connections and a corresponding port connection in the second set of port connections. For example, each DFE circuit associated with a DQ line is configured to make corrections in one or more DQ signals

U.S. Patent No. 10,025,731 Domestic Industry Claim Chart (Samsung)

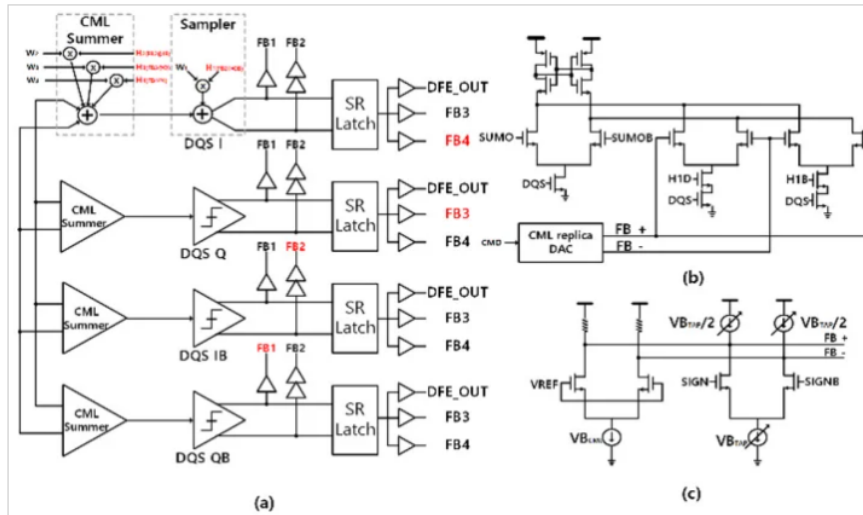
Claim 1	Evidence of Use																		
	transmitted between a respective port connection of the first set of port connections connected to the edge connector (e.g., a data port on the host side) and a corresponding port connection for a DDR5 memory device in the second set of port connections (e.g., a data port on the DRAM). For example, Samsung's presentation at Hot Chips 2021 shows that DFE coupled with variable Rx generates delayed versions of the incoming DQ signals, which are used to make corrections to the incoming DQ signals.  Faster Transfer Speed By newly adopted equalization schemes DFE Circuit • Decision feedback equalizer (DFE) with loop-unrolling • Shortened data path with un-matched data and strobe (DQ-DQS) • Equalizer scheme with per-pin VrefDQ calibration Test Result <table border="1"> <thead> <tr> <th>VDDQ</th> <th>OPS</th> </tr> </thead> <tbody> <tr> <td>1.300V</td> <td>250 OPS</td> </tr> <tr> <td>1.250V</td> <td>250 OPS</td> </tr> <tr> <td>1.200V</td> <td>250 OPS</td> </tr> <tr> <td>1.150V</td> <td>250 OPS</td> </tr> <tr> <td>1.100V</td> <td>270 OPS</td> </tr> <tr> <td>1.050V</td> <td>270 OPS</td> </tr> <tr> <td>1.000V</td> <td>290 OPS</td> </tr> <tr> <td>950.0mV</td> <td>310 OPS</td> </tr> </tbody> </table> Ex. CC-105 (Samsung Hot Chips 2021 Presentation) at 5. See also, e.g., Ex. CC-106 (Samsung's massive DDR5 512GB RAM modules) at 4 ("Samsung has also upgraded its new RAM with a new resolution (DFE) rating to improve signal stability."). - Ex. CC-107 (All About Circuits ISSCC 2024) at 4:	VDDQ	OPS	1.300V	250 OPS	1.250V	250 OPS	1.200V	250 OPS	1.150V	250 OPS	1.100V	270 OPS	1.050V	270 OPS	1.000V	290 OPS	950.0mV	310 OPS
VDDQ	OPS																		
1.300V	250 OPS																		
1.250V	250 OPS																		
1.200V	250 OPS																		
1.150V	250 OPS																		
1.100V	270 OPS																		
1.050V	270 OPS																		
1.000V	290 OPS																		
950.0mV	310 OPS																		

U.S. Patent No. 10,025,731 Domestic Industry Claim Chart (Samsung)

Claim 1

Evidence of Use

Samsung's proposed architecture adds to conventional DFE circuits with a four-tap system—an alternative to conventional two-tap DFE. In a DFE circuit, the taps are fed back to the input and summed. The four-tap directly feeds tap one back to minimize feedback delay. The second, third, and fourth taps use current mode logic (CML) summation to further enhance the symbol accuracy.



(a) Four-tap DFE. (b) Tap one directly feeds the sampler. (c) Taps two through four into the CML circuit.

The DFE operates along with automatic-offset-voltage-calibration circuitry in the DQ buffers. The calibration circuit compensates for offset by using four paths for four phases of the operation and calibrates based on a direct majority voting of the four path outputs. The result is the ability to reliably operate at 8 Gb/s or faster.

- Ex. CC-108 (FAC Ex. 10, 25-cv-557 (EDTX) at 19–20 (“Recent Evolution in the DRAM Interface: Mile-Markers Along Memory Lane”: “[T]he DDR5 die architecture introduces several circuit-based features to address the challenging signaling environment. Perhaps the

U.S. Patent No. 10,025,731 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	most profound enhancement is the integration of four taps of decision feedback equalization (DFE) at the DRAM data path input. While several equalization methods might have been adopted, DFE was chosen based on the nature and severity of the multidrop interconnect previously referenced.”). • Ex. CC-58 (Kim 2020) at 168 (annotated) (a SK Hynix article illustrating basic functional blocks of a DDR5 SDRAM, which is believed to also exist in a Samsung DDR5 SDRAM):

Claim 1

Evidence of Use

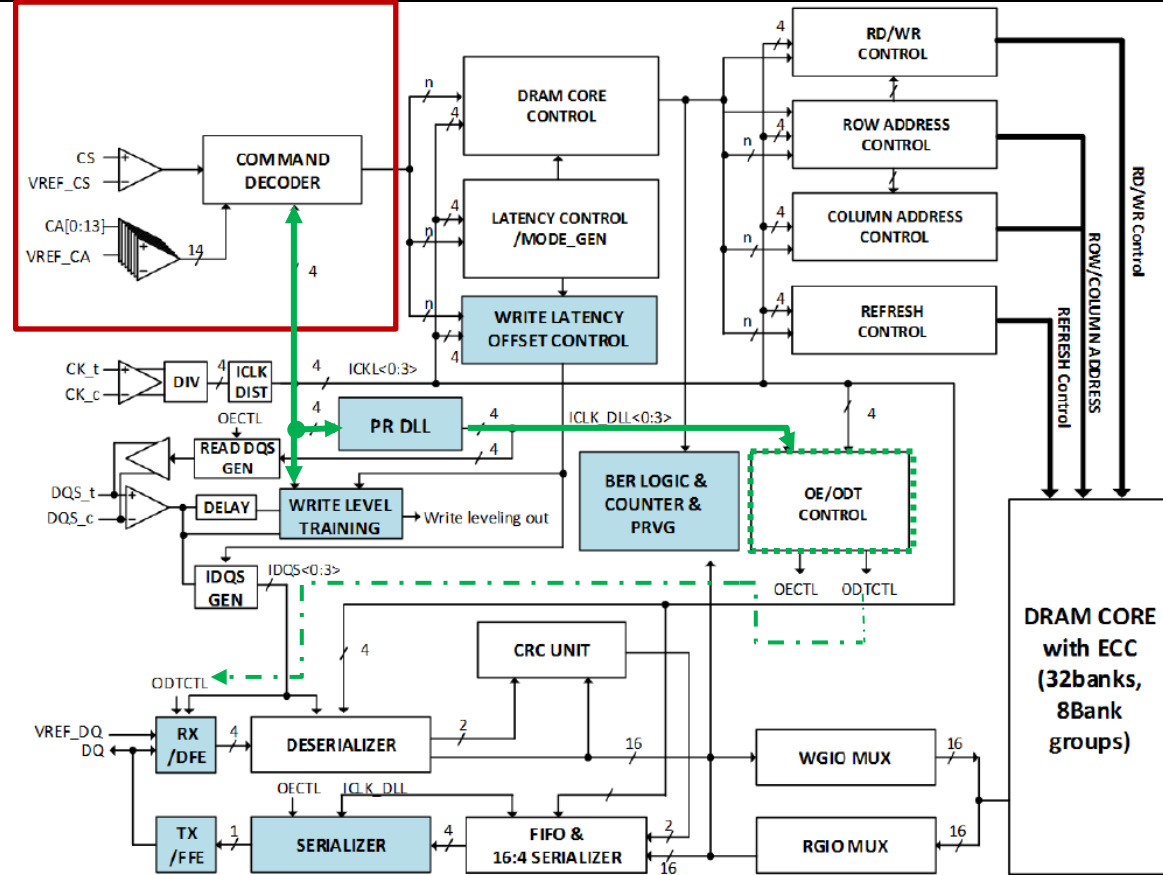


Fig. 2. DDR5 functional block diagram.

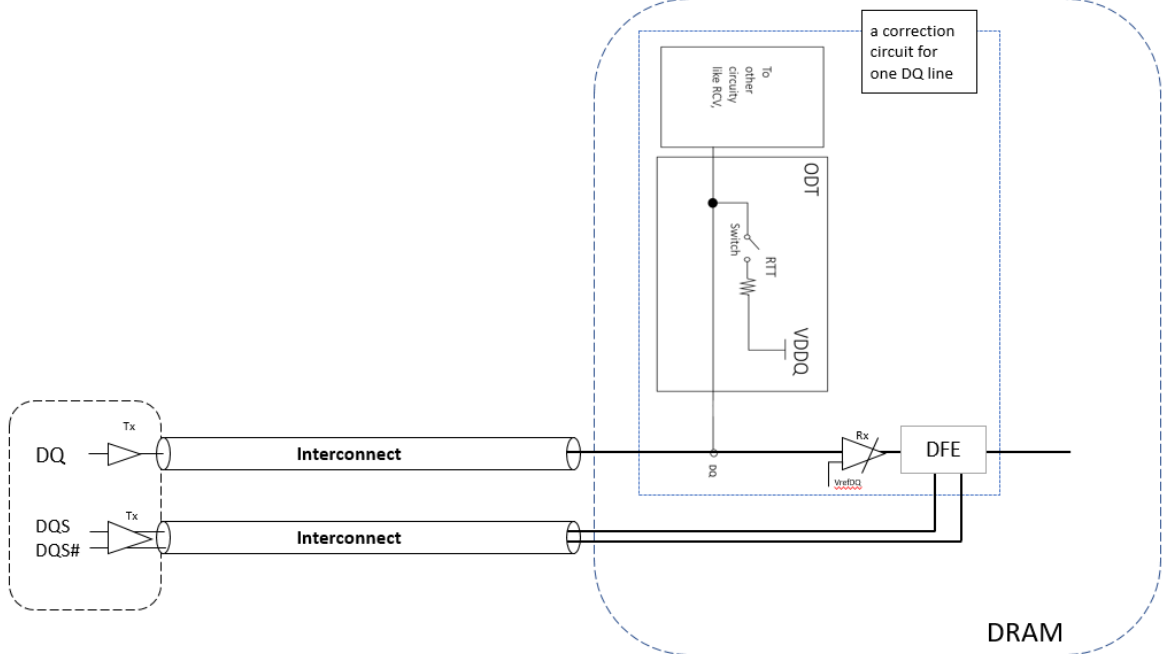
This claim element is literally present, as described herein, but is also satisfied under the doctrine of equivalents, and the Samsung Practicing '731 Products are manufactured in a manner that is not

U.S. Patent No. 10,025,731 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	substantially different from the requirements of this limitation because they achieve substantially or exactly the same function, in substantially or exactly the same way as set forth in the claim element (as described herein), and achieve substantially or exactly the same result.

Claim 1	Evidence of Use
[1.f] the each correction circuit of the set of correction circuits including at least one programmable impedance matching circuit; and,	<i>The each correction circuit of the set of correction circuits include[s] at least one programmable impedance matching circuit</i> such as the ODT/RTT circuit for a DQ line. <i>See supra</i> limitations [1.a]-[1.e]. For example, as stated in Ex. CC-10 (Intellect Partners DD5’s Secret Weapon_ODT), “[b]y placing a termination resistor that matches the transmission line’s impedance right on the memory chip, on-die termination minimizes the possibility of signal reflections. Therefore, ODT is a crucial component for high-speed DDR5 memory systems since it aids in enhancing signal quality, decreasing signal ringing, and eventually allowing for higher data transfer speeds with less signal deterioration.” <i>Id.</i> at 2. As illustrated above and below, each DQ includes a correction circuit comprising a DDR5-specific DFE circuit and an ODT/RTT circuit:

U.S. Patent No. 10,025,731 Domestic Industry Claim Chart (Samsung)

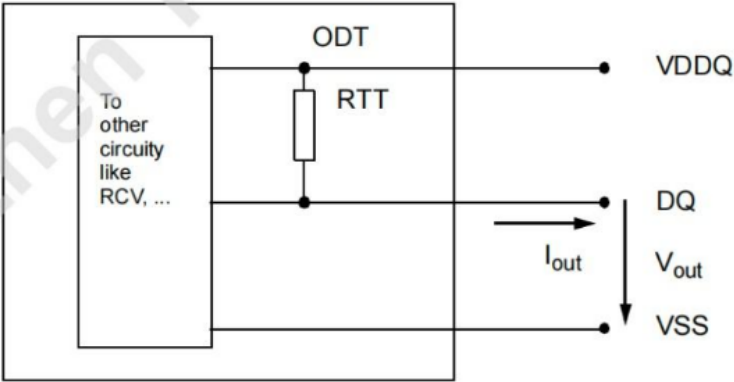
Claim 1	Evidence of Use
	 Each correction circuit includes the DFE equalization circuitry to make corrections to one input signal on one DQ pin and the programmable RTT connected to DQ line. The ODT/RTT circuit is a programmable impedance matching circuit:

U.S. Patent No. 10,025,731 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	[Source: DDR5 Standard [JEDEC JESD79-5B_v1.20] Page 346 of 502] Reproduced from Ex. CC-10 (Intellect Partners DD5's Secret Weapon_ODT) which in turn cites from JEDEC JESD79-5B_v1.20. The RTT is programmable depending on the values in the mode registers. <i>See, e.g.</i>, Ex. CC-89 (Samsung DDR5 RDIMM Datasheet) at 73–75, [Table 48] Basic IDD, IDDQ and IPP Measurement Conditions (showing “Output Buffer and RTT: Enabled in Mode Registers2”); <i>see also id.</i> at Note 2 (showing that RTT_Nom (which include RTT_Nom_RD and RTT_Nom_WR), RTT_WR, and RTT_Park are programmed in MR35 OP[5:0], MR34 OP[5:3], and MR34 OP[2:0], respectively): <pre> 2) Output Buffer Enable - set MR5 OP[0] = 0] : Qoff = Output buffer enabled - set MR5 OP[2:1] = 00]: Pull-Up Output Driver Impedance Control = RZQ/7 - set MR5 OP[7:6] = 00]: Pull-Down Output Driver Impedance Control = RZQ/7 RTT_Nom enable - set MR35 OP[5:0] = 110110: RTT_NOM_WR = RTT_NOM_RD = RZQ/6 RTT_WR enable - set MR34 OP[5:3] = 010 RTT_WR = RZQ/2 RTT_PARK disable - set MR34 OP[2:0] = 000 </pre> Ex. CC-109 (Intel Dynamic ODT in DDR4) at 154 (emphasis added) (“In DDR4, in addition to the Rtt_nom and Rtt_wr values, which are applied during read and write respectively, a third option called Rtt_park is available. When Rtt_park is enabled, a selected termination value is set in the

U.S. Patent No. 10,025,731 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use																																																																																			
	DRAM when ODT is driven low. Refer to the DDR4 JEDEC* specification or your memory vendor data sheet for details about available termination values and functional description for <i>dynamic ODT</i> in DDR4 devices.”). Ex. CC-59 (JEDEC D5 Chapter5) at 2 (explaining that per JEDEC DDR5 specification, there are five RTT values for data in the order of priority: (1) data termination disable, (2) RTT_WR, (3) RTT_NOM_RD, (4) RTT_NOM_WR, and (5) RTT_PARK; and RTT_PARK is suitable for multi-rank systems so that, for example, when the system memory controller is writing to Rank0, the other ranks need not be placed in Hi-Z, thus improving the DRAM efficiency); <i>id.</i> at 3:																																																																																			
	Table 177 — Termination State Table <table><tr><th rowspan="2">Command</th><th colspan="4">Mode Register Configuration Settings</th><th colspan="2">Results</th><th rowspan="2">Note</th></tr><tr><th>RTT_PARK</th><th>RTT_WR</th><th>RTT_NOM_WR</th><th>RTT_NOM_RD</th><th>Target DRAM Term</th><th>Non-Target DRAM Term</th></tr><tr><td>ANY</td><td colspan="4">Disabled</td><td colspan="2">HI-Z (ODT OFF)</td><td>3</td></tr><tr><td>Any Non-Term CMD</td><td>Enabled</td><td colspan="3">Don't Care</td><td>RTT_PARK</td><td>RTT_PARK</td><td>4</td></tr><tr><td rowspan="7">WR</td><td colspan="3">Disabled</td><td rowspan="7">Don't Care</td><td>HI-Z (ODT OFF)</td><td>HI-Z (ODT OFF)</td><td rowspan="7">2</td></tr><tr><td colspan="2">Disabled</td><td>Enabled</td><td>HI-Z (ODT OFF)</td><td>RTT_NOM_WR</td></tr><tr><td>Disabled</td><td>Enabled</td><td>Disabled</td><td>RTT_WR</td><td>HI-Z (ODT OFF)</td></tr><tr><td>Don't Care</td><td colspan="2">Enabled</td><td>RTT_WR</td><td>RTT_NOM_WR</td></tr><tr><td colspan="2">Enabled</td><td>Disabled</td><td>RTT_WR</td><td>HI-Z (ODT OFF)</td></tr><tr><td>Enabled</td><td>Disabled</td><td>Enabled</td><td>RTT_PARK</td><td>RTT_NOM_WR</td></tr><tr><td>Enabled</td><td colspan="2">Disabled</td><td>RTT_PARK</td><td>HI-Z (ODT OFF)</td></tr><tr><td rowspan="2">RD/MRR</td><td>Enabled</td><td colspan="2">Don't Care</td><td>Disabled</td><td>HI-Z (ODT OFF)</td><td>HI-Z (ODT OFF)</td><td>1,5</td></tr><tr><td colspan="3">Don't Care</td><td>Enabled</td><td>HI-Z (ODT OFF)</td><td>RTT_NOM_RD</td><td>1</td></tr></table> NOTE 1 When read command is executed, DRAM termination state of target rank will be Hi-Z for defined period independent of MR setting of RTT_PARK/RTT_NOM_RD/RTT_NOM_WR. NOTE 2 If RTT_WR is enabled, RTT_WR will be activated by Write command for defined period time independent of MR setting of RTT_PARK/RTT_NOM. NOTE 3 If all RTT configs are disabled, ODT receiver power will be turned off to save power. NOTE 4 If RTT_PARK is enabled, DRAM RTT_PARK termination will be enabled while WR/RD/MRR are not being executed. NOTE 5 When a Non-Target ODT command is executed and the RTT_NOM_WR or RTT_NOM_RD is disabled, the non-target rank will be Hi-Z for a defined period, independent of the MR setting of RTT_PARK.	Command	Mode Register Configuration Settings				Results		Note	RTT_PARK	RTT_WR	RTT_NOM_WR	RTT_NOM_RD	Target DRAM Term	Non-Target DRAM Term	ANY	Disabled				HI-Z (ODT OFF)		3	Any Non-Term CMD	Enabled	Don't Care			RTT_PARK	RTT_PARK	4	WR	Disabled			Don't Care	HI-Z (ODT OFF)	HI-Z (ODT OFF)	2	Disabled		Enabled	HI-Z (ODT OFF)	RTT_NOM_WR	Disabled	Enabled	Disabled	RTT_WR	HI-Z (ODT OFF)	Don't Care	Enabled		RTT_WR	RTT_NOM_WR	Enabled		Disabled	RTT_WR	HI-Z (ODT OFF)	Enabled	Disabled	Enabled	RTT_PARK	RTT_NOM_WR	Enabled	Disabled		RTT_PARK	HI-Z (ODT OFF)	RD/MRR	Enabled	Don't Care		Disabled	HI-Z (ODT OFF)	HI-Z (ODT OFF)	1,5	Don't Care			Enabled	HI-Z (ODT OFF)	RTT_NOM_RD	1
Command	Mode Register Configuration Settings				Results		Note																																																																													
	RTT_PARK	RTT_WR	RTT_NOM_WR	RTT_NOM_RD	Target DRAM Term	Non-Target DRAM Term																																																																														
ANY	Disabled				HI-Z (ODT OFF)		3																																																																													
Any Non-Term CMD	Enabled	Don't Care			RTT_PARK	RTT_PARK	4																																																																													
WR	Disabled			Don't Care	HI-Z (ODT OFF)	HI-Z (ODT OFF)	2																																																																													
	Disabled		Enabled		HI-Z (ODT OFF)	RTT_NOM_WR																																																																														
	Disabled	Enabled	Disabled		RTT_WR	HI-Z (ODT OFF)																																																																														
	Don't Care	Enabled			RTT_WR	RTT_NOM_WR																																																																														
	Enabled		Disabled		RTT_WR	HI-Z (ODT OFF)																																																																														
	Enabled	Disabled	Enabled		RTT_PARK	RTT_NOM_WR																																																																														
	Enabled	Disabled			RTT_PARK	HI-Z (ODT OFF)																																																																														
RD/MRR	Enabled	Don't Care		Disabled	HI-Z (ODT OFF)	HI-Z (ODT OFF)	1,5																																																																													
	Don't Care			Enabled	HI-Z (ODT OFF)	RTT_NOM_RD	1																																																																													

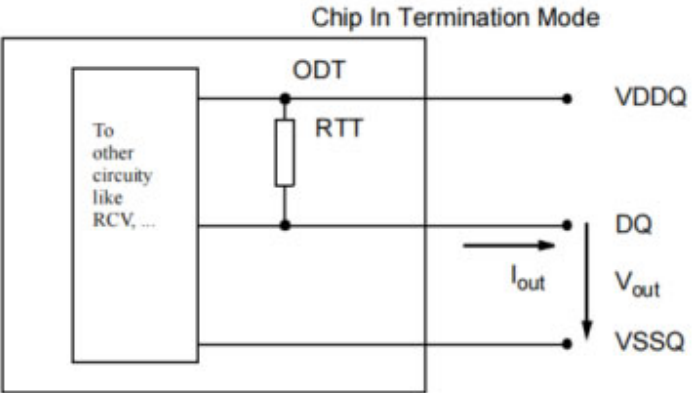
Claim 1	Evidence of Use
	Chip In Termination Mode  Figure 171 — On Die Termination 知乎 @sakui <i>See also, e.g.,</i> Ex. CC-60 (CSDN Blog, JESD79-5) at 1–2: - RTT_WR: The rank that is being written to provide termination and adjusts timing based on ODT Control Mode Register settings. - RTT_NOM_RD: DRAM turns ON RTT_NOM_RD if it sees CS asserted during the second pulse of the READ command (except when ODT is disabled by MR35:OP[5:3]). - RTT_NOM_WR: DRAM turns ON RTT_NOM_WR if it sees a CS asserted during the second pulse of the WRITE command (except when ODT is disabled by MR35:OP[2:0]). - RTT_PARK: Default parked value set via MR34:OP[2:0] and is to be enabled when a READ or WRITE is not active. - DQS_RTT_PARK: Default parked value set for DQS via MR33:OP[5:3] and is to be enabled when a READ is not active - Data Termination Disable: DRAM driving data upon receiving READ command disables the termination after RL-1 and stays off for a duration of BL/2. - Strobe Termination Disable: DRAM driving strobe upon receiving READ command disables the termination after RL-1-tRPST and stays off for a duration of BL/2+tRPST.

U.S. Patent No. 10,025,731 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	Those RTT values have priority as follows: 1 Data Termination Disable & Strobe Termination Disable 2 RTT_WR 3 RTT_NOM_RD 4 RTT_NOM_WR 5 RTT_PARK This means that if there is a WRITE command, then the DRAM turns on RTT_WR, not RTT_NOM_WR or RTT_NOM_RD, and also if there is a READ command, then the DRAM disables data termination and goes into Driving mode. If during the second pulse of a READ or WRITE command, a CS enable is sent, then Non-Target ODT is enabled and the appropriate RTT_NOM_RD or RTT_NOM_WR is enabled for the non-target rank. This provides additional and potentially different termination options for the other ranks on the channel. Application: Controller can control each RTT condition with WR/RD command and ODT pin - RTT_WR: The rank that is being written to provide termination regardless of ODT pin status (either HIGH or LOW) - RTT_NOM: DRAM turns ON RTT_NOM if it sees ODT asserted (except ODT is disabled by MR1). - RTT_PARK: Default parked value set via MR5 to be enabled and ODT pin is driven LOW. - Data Termination Disable: DRAM driving data upon receiving READ command disables the termination after RL-X and stays off for a duration of BL/2 + X clock cycles. X is 2 for 1tCK and 3 for 2tCK preamble mode. - The Termination State Table is shown in Table 72. Those RTT values have priority as following. 1. Data Termination Disable 2. RTT_WR 3. RTT_NOM 4. RTT_PARK which means if there is WRITE command along with ODT pin HIGH, then DRAM turns on RTT_WR not RTT_NOM, and also if there is READ command, then DRAM disables data termination regardless of ODT pin and goes into Driving mode.

U.S. Patent No. 10,025,731 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use																																																																																									
	Table 384 — Termination State Table <table><tr><th rowspan="2">Command</th><th colspan="4">Mode Register Configuration Settings</th><th colspan="2">Results</th><th rowspan="2">Note</th></tr><tr><th>RTT_PA</th><th>RTT_W</th><th>RTT_NOM_</th><th>RTT_NOM_</th><th>Target DRAM Term</th><th>Non-Target DRAM</th></tr><tr><td>ANY</td><td colspan="4">Disabled</td><td colspan="2">HI-Z (ODT OFF)</td><td>3</td></tr><tr><td>Any Non-</td><td>Enabled</td><td colspan="3">Don't Care</td><td>RTT_PARK</td><td>RTT_PARK</td><td>4</td></tr><tr><td rowspan="7">WR</td><td colspan="3">Disabled</td><td rowspan="7">Don't Care</td><td>HI-Z (ODT OFF)</td><td>HI-Z (ODT OFF)</td><td></td></tr><tr><td colspan="2">Disabled</td><td>Enabled</td><td>HI-Z (ODT OFF)</td><td>RTT_NOM_WR</td><td></td></tr><tr><td>Disabled</td><td>Enabled</td><td>Disabled</td><td>RTT_WR</td><td>HI-Z (ODT OFF)</td><td></td></tr><tr><td>Don't</td><td colspan="2">Enabled</td><td>RTT_WR</td><td>RTT_NOM_WR</td><td>2</td></tr><tr><td colspan="2">Enabled</td><td>Disabled</td><td>RTT_WR</td><td>HI-Z (ODT OFF)</td><td>2,5</td></tr><tr><td>Enabled</td><td>Dis-</td><td>Enabled</td><td>RTT_PARK</td><td>RTT_NOM_WR</td><td></td></tr><tr><td>Enabled</td><td colspan="2">Disabled</td><td>RTT_PARK</td><td>HI-Z (ODT OFF)</td><td>5</td></tr><tr><td rowspan="2">RD/MRR</td><td>Enabled</td><td colspan="2">Don't Care</td><td>Disabled</td><td>HI-Z (ODT OFF)</td><td>HI-Z (ODT OFF)</td><td>1,5</td></tr><tr><td colspan="3">Don't Care</td><td>Enabled</td><td>HI-Z (ODT OFF)</td><td>RTT_NOM_RD</td><td>1</td></tr></table> NOTE 1 When read command is executed, DRAM termination state of target rank will be Hi-Z for defined period independent of MR setting of RTT_PARK/RTT_NOM_RD/RTT_NOM_WR. NOTE 2 If RTT_WR is enabled, RTT_WR will be activated by Write command for defined period time independent of MR setting of RTT_PARK /RTT_NOM. NOTE 3 If all RTT configs are disabled, ODT receiver power will be turned off to save power. NOTE 4 If RTT_PARK is enabled, DRAM RTT_PARK termination will be enabled while WR/RD/MRR are not being executed. NOTE 5 When a Non-Target ODT command is executed and the RTT_NOM_WR or RTT_NOM_RD is disabled, the DRAM termination state of the non-target rank will be Hi-Z for a defined period, independent of the MR setting of RTT_PARK.	Command	Mode Register Configuration Settings				Results		Note	RTT_PA	RTT_W	RTT_NOM_	RTT_NOM_	Target DRAM Term	Non-Target DRAM	ANY	Disabled				HI-Z (ODT OFF)		3	Any Non-	Enabled	Don't Care			RTT_PARK	RTT_PARK	4	WR	Disabled			Don't Care	HI-Z (ODT OFF)	HI-Z (ODT OFF)		Disabled		Enabled	HI-Z (ODT OFF)	RTT_NOM_WR		Disabled	Enabled	Disabled	RTT_WR	HI-Z (ODT OFF)		Don't	Enabled		RTT_WR	RTT_NOM_WR	2	Enabled		Disabled	RTT_WR	HI-Z (ODT OFF)	2,5	Enabled	Dis-	Enabled	RTT_PARK	RTT_NOM_WR		Enabled	Disabled		RTT_PARK	HI-Z (ODT OFF)	5	RD/MRR	Enabled	Don't Care		Disabled	HI-Z (ODT OFF)	HI-Z (ODT OFF)	1,5	Don't Care			Enabled	HI-Z (ODT OFF)	RTT_NOM_RD	1
Command	Mode Register Configuration Settings				Results		Note																																																																																			
	RTT_PA	RTT_W	RTT_NOM_	RTT_NOM_	Target DRAM Term	Non-Target DRAM																																																																																				
ANY	Disabled				HI-Z (ODT OFF)		3																																																																																			
Any Non-	Enabled	Don't Care			RTT_PARK	RTT_PARK	4																																																																																			
WR	Disabled			Don't Care	HI-Z (ODT OFF)	HI-Z (ODT OFF)																																																																																				
	Disabled		Enabled		HI-Z (ODT OFF)	RTT_NOM_WR																																																																																				
	Disabled	Enabled	Disabled		RTT_WR	HI-Z (ODT OFF)																																																																																				
	Don't	Enabled			RTT_WR	RTT_NOM_WR	2																																																																																			
	Enabled		Disabled		RTT_WR	HI-Z (ODT OFF)	2,5																																																																																			
	Enabled	Dis-	Enabled		RTT_PARK	RTT_NOM_WR																																																																																				
	Enabled	Disabled			RTT_PARK	HI-Z (ODT OFF)	5																																																																																			
RD/MRR	Enabled	Don't Care		Disabled	HI-Z (ODT OFF)	HI-Z (ODT OFF)	1,5																																																																																			
	Don't Care			Enabled	HI-Z (ODT OFF)	RTT_NOM_RD	1																																																																																			

Claim 1	Evidence of Use
	On-Die Termination effective resistance R_{TT} is defined by MR bits. ODT is applied to the DQ, DM, DQS_T/DQS_C and TDQS_T/TDQS_C (x8 devices only) pins. A functional representation of the on-die termination is shown in Figure 153. $R_{TT} = \frac{V_{DDQ} - V_{out}}{ I_{out} }$  Figure 153 — On Die Termination On die termination effective R_{tt} values supported are 240, 120, 80, 60, 48, 40, 34 ohms.

U.S. Patent No. 10,025,731 Domestic Industry Claim Chart (Samsung)

Table 385 — ODT Electrical Characteristics RZQ=240Ω +/-1% Entire Temperature Operation Range; after Proper ZQ Calibration

RTT	Vout	Min	Nom	Max	Unit	NOTE
240Ω	VOLdc= 0.5* VDDQ	0.9	1	1.25	RZQ	1,2,3
	VOMdc= 0.8* VDDQ	0.9	1	1.1	RZQ	1,2,3
	VOHdc= 0.95* VDDQ	0.8	1	1.1	RZQ	1,2,3
120Ω	VOLdc= 0.5* VDDQ	0.9	1	1.25	RZQ/2	1,2,3
	VOMdc= 0.8* VDDQ	0.9	1	1.1	RZQ/2	1,2,3
	VOHdc= 0.95* VDDQ	0.8	1	1.1	RZQ/2	1,2,3
80Ω	VOLdc= 0.5* VDDQ	0.9	1	1.25	RZQ/3	1,2,3
	VOMdc= 0.8* VDDQ	0.9	1	1.1	RZQ/3	1,2,3
	VOHdc= 0.95* VDDQ	0.8	1	1.1	RZQ/3	1,2,3
60Ω	VOLdc= 0.5* VDDQ	0.9	1	1.25	RZQ/4	1,2,3
	VOMdc= 0.8* VDDQ	0.9	1	1.1	RZQ/4	1,2,3
	VOHdc= 0.95* VDDQ	0.8	1	1.1	RZQ/4	1,2,3
48Ω	VOLdc= 0.5* VDDQ	0.9	1	1.25	RZQ/5	1,2,3
	VOMdc= 0.8* VDDQ	0.9	1	1.1	RZQ/5	1,2,3
	VOHdc= 0.95* VDDQ	0.8	1	1.1	RZQ/5	1,2,3
40Ω	VOLdc= 0.5* VDDQ	0.9	1	1.25	RZQ/6	1,2,3
	VOMdc= 0.8* VDDQ	0.9	1	1.1	RZQ/6	1,2,3
	VOHdc= 0.95* VDDQ	0.8	1	1.1	RZQ/6	1,2,3
34Ω	VOLdc= 0.5* VDDQ	0.9	1	1.25	RZQ/7	1,2,3
	VOMdc= 0.8* VDDQ	0.9	1	1.1	RZQ/7	1,2,3
	VOHdc= 0.95* VDDQ	0.8	1	1.1	RZQ/7	1,2,3
DQ-DQMismatch within byte	VOMdc = 0.8* VDDQ	0	-	8	%	1,2,4,5,6

NOTE 1 The tolerance limits are specified after calibration with stable voltage and temperature. For the behavior of the tolerance limits if temperature or voltage changes after calibration, see Chapter 6.

NOTE 2 Pull-up ODT resistors are recommended to be calibrated at 0.8*VDDQ. Other calibration schemes may be used to achieve the linearity spec shown in the table, e.g., calibration at 0.5*VDDQ and 0.95*VDDQ.

NOTE 3 The tolerance limits are specified under the condition that VDDQ=VDD and VSSQ=VSS

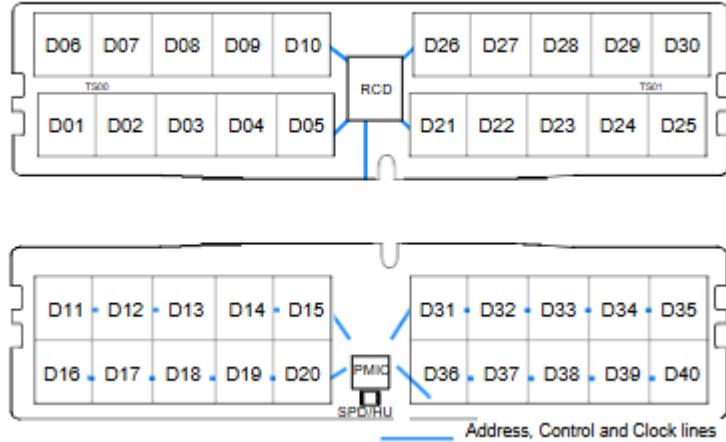
NOTE 4 DQ to DQ mismatch within byte variation for a given component including DQS_T and DQS_C (characterized)

U.S. Patent No. 10,025,731 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	This claim element is literally present, as described herein, but is also satisfied under the doctrine of equivalents, and the Samsung Practicing '731 Products are manufactured in a manner that is not substantially different from the requirements of this limitation because they achieve substantially or exactly the same function, in substantially or exactly the same way as set forth in the claim element (as described herein), and achieve substantially or exactly the same result.

Claim 1	Evidence of Use
[1.g] control circuitry configured to receive the address and control signals from the system memory controller and to control the plurality of memory devices in response to the address and control signals,	Samsung's DDR5 RDIMM and MRDIMMs each include control circuitry (e.g., the control/address circuit on SDRAMs and where available, the RCD or MRCD) configured to receive the address and control signals from the system memory controller and to control the plurality of memory devices in response to the address and control signals. See <i>supra</i> limitations [1.b]-[1.f]. For example, in RDIMMs, as Samsung's DDR5 RDIMMs show, the RCD receives via the edge connection, control and address signals from the memory controller:

U.S. Patent No. 10,025,731 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	 Ex. CC-89 (Samsung DDR5 RDIMM Datasheet) at 33; <i>see also id.</i> at 29, 25, 21. As the figures above and below also show, the RCD takes a copy of the C/A signals received over seven C/A[6:0] edge pins and two CS[1:0] edge pins for each channel and over two cycles to generate two corresponding copies of the control/address and CS signals (one for front side and one for backside) for output to the plurality of memory devices (<i>e.g.</i>, via the control/address signal receiving and processing circuit on the SDRAMs). Each copy of the C/A signals is output over 14 C/A output to a corresponding SDRAM:

U.S. Patent No. 10,025,731 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	DCA[06:00]_A ——— QACA [13:00]_A -> CA [13:00] : SDRAMs D [10:01] DCS0_A_n ——— QACS0_A->CS_n:SDRAMsD[05:01] DCS0_A->CS_n:SDRAMsD[10:06] DCS1_A_n ——— DCA [06:00]_B ——— QACA[13:00]_B -> CA[13:00]: SDRAMs D[20:11] DCS0_B_n ——— R QACS0_B -> CS_n : SDRAMs D[15:11] E QACS0_B -> CS_n : SDRAMs D[20:16] G DCS1_B_n ——— I DCK_t / DCK_c ——— T QACK_A_t->CK_t: SDRAMs D[05:01] / QACK_A_c->CK_c: SDRAMs D [05:01] E QCCCK_A_t->CK_t: SDRAMs D[10:06] / QCCCK_A_c->CK_c: SDRAMs D [10:06] R QACK_B_t->CK_t: SDRAMs D[15:11] / QACK_B_c->CK_c: SDRAMs D [15:11] QCCCK_B_t->CK_t: SDRAMs D[20:16] / QCCCK_B_c->CK_c: SDRAMs D [20:16] QLBD ——— DLBD_A->LBDQ: SDRAMsD[10:01] DLBD_B->LBDQ: SDRAMsD[20:11] QLBS ——— DLBS_A->LBDQS: SDRAMsD[10:01] DLBS_B->LBDQS: SDRAMs D[20:11] DRST_n ——— QRT_A_n->RESET_n: SDRAMsD[10:01] QRT_B_n -> RESET_n : SDRAMs D[20:11] ALERT_n ——— DERROR_IN_A_n->ALERT_n : SDRAMs D [10:01] DERROR_IN_B_n->ALERT_n : SDRAMs D [20:11] Ex. CC-89 (Samsung DDR5 RDIMM Datasheet) at 33; <i>see also id.</i> at 29, 25, 21.

Claim 1

Evidence of Use

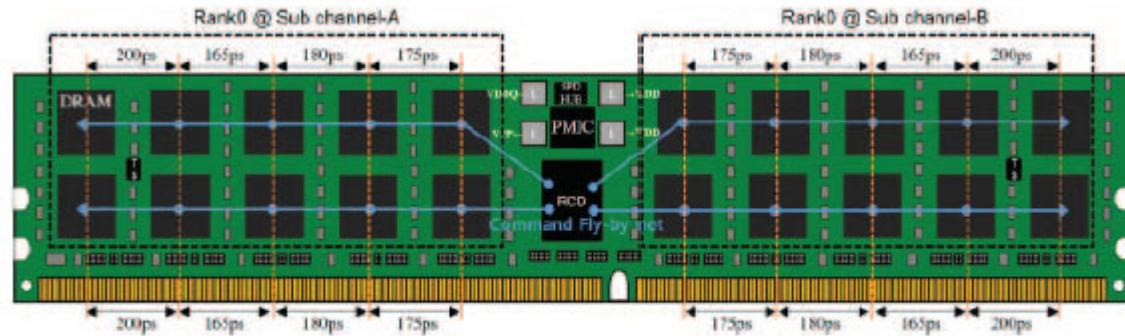


Figure 3. The operation delay due to fly-by topology of command and address nets in DDR5 RDIMM 2Rx4.

Ex. CC-45 (Na 2022) at 931, Fig. 3 (showing RCD outputting control/address signals to SDRAMs).

See also Ex. CC-96 (Rambus DDR5 RCD) at 1 (emphasis added and annotated) (Rambus is a supplier of Samsung DDR5 RDIMM.) (“*The Rambus DDR5 RCD provides Command/Address (CA) and clocks to the DDR5 memory devices in RDIMMs. . . . The RCD distributes C/A signals and clocks for each subchannel, rank and nibble.*”):

U.S. Patent No. 10,025,731 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	DDR5 RCD High-Level Functional Block Diagram Ex. CC-48 (Montage M88DR5RCD01) at 1 (Montage is also an RCD supplier for Samsung’s DDR5 RDIMMs.) (“It is a two-channel RCD designed for driving addresses and controlling nets between the host controller and the DDR5 SDRAMs.”):

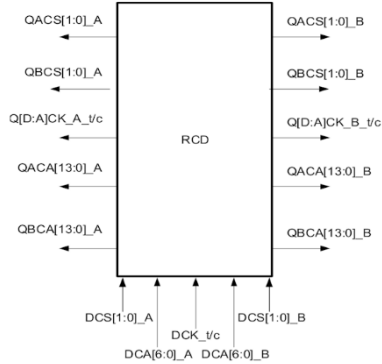
U.S. Patent No. 10,025,731 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	The diagram illustrates a memory module architecture. At the top, there is a green rectangular module containing a grid of SDRAM chips. The grid is 2 rows by 12 columns. The top row has 10 SDRAM chips, followed by a central RCD chip, and then 2 more SDRAM chips. The bottom row has 10 SDRAM chips, followed by the same central RCD chip, and then 2 more SDRAM chips. Below the module, a bus labeled 'CMD/ADDR/CTRL/CLK' connects to a 'DDR5 Memory Controller' block. Ex. CC-87 (Montage M88DR5RCD02) at 1 (“M88DR5RCD02 is a Gen2 DDR5 Registering Clock Driver (RCD02) with a maximum data rate of 5600MT/s. Compliant with the latest JEDEC DDR5RCD02 specification, the RCD02 contains two separate channels and supports 1.1V VDD supply.”); Ex. CC-97 (Montage M88DR5RCD03) at 1 (“M88DR5RCD03 is a Gen3 DDR5 Registering Clock Driver (RCD03) with a maximum data rate of 6400MT/s. Compliant with the latest JEDEC DDR5RCD03 specification, the RCD03 contains two separate channels and supports 1.1V VDD supply.”); Ex. CC-98 (Montage M88DR5RCD04) at 1 (“M88DR5RCD04 is a Gen4 DDR5 Registering Clock Driver (RCD04) with a maximum data rate of 7200MT/s. Compliant with JEDEC DDR5RCD04 specification, the RCD04 contains two separate channels and supports 1.1V VDD supply. Designed for use on DDR5 RDIMMs, the chip can help the upcoming server systems address performance bottlenecks and provide critical support for demanding workloads” such as artificial intelligence, machine learning, and high-performance computing.):

Claim 1	Evidence of Use
	The diagram illustrates a system architecture. At the bottom is a grey rounded rectangle labeled "DDR5 Memory Controller". A vertical line connects it to a black square chip labeled "DDR5 RCD02". This chip is connected to a green memory module. The module contains two rows of green chips, each labeled "SDRAM". A small black chip with a white "M" logo is also visible on the module. Ex. CC-52 (Gen 1 RCD Renesas 5RCD0148) at 1 (emphasis added) (Gen 1 RCD by Renesas, another supplier to Samsung) (“The 5RCD0148HC2 (RCD) is a registering clock driver used on DDR5 RDIMMs and LRDIMMs. It supports DDR5 server speeds up to 4800 MT/s. <i>Its primary function is to buffer the Command Address (CA) bus, chip selects, and clock between the host controller and the DRAMs.</i> It also creates a BCOM bus to control the data buffers for LRDIMMs. The 5RCD0148HC2 contains two separate channels with some common logic such as clocking, but otherwise operate independently of each other. Each channel has a 7-bit double data rate CA bus input, a single parity input, two chip-select inputs, produces two copies of 14-bit single data rate CA bus outputs and two copies of the chip-select outputs. The RCD has a common clock input and PLL but produces separate clock outputs to the DRAM channels.”): The diagram shows a central block labeled "RCD". At the bottom, three arrows point up into the block, labeled "DOCA[6:0]", "CLOCK", and "D1CA[6:0]". On the left side, four arrows point out of the block, labeled "BCOM0 BUS", "Y0 CLOCKS", "Q0ACA[13:0]", and "Q0BCA[13:0]". On the right side, four arrows point out of the block, labeled "BCOM1 BUS", "Y1 CLOCKS", "Q1ACA[13:0]", and "Q1BCA[13:0]".

Claim 1	Evidence of Use
	Ex. CC-51 (Gen 2 RCD Renesas RG5R256) at 1 (Gen 2 RCD): Ex. CC-99 (Gen 3 RCD Renesas RG5R364) at 1 (Gen 3 RCD): Ex. CC-100 (Gen 4 RCD Renesas RRG5004) at 1 (Gen 4 RCD by Renesas):

U.S. Patent No. 10,025,731 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	Block Diagram  The C/A and CS signals the SDRAM receive from the memory controller and RCD/MRCD control the operation of the memory device. <i>See</i> Ex. CC-46 (DDR5 Command Truth Table) at 1 (showing that commands for Activate, Read, and Write are composed of C/A[0:13] and CS signals):

U.S. Patent No. 10,025,731 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use																		
	Function	Abbreviation	CS_n	CA Pins													NOTES		
				CA0	CA1	CA2	CA3	CA4	CA5	CA6	CA7	CA8	CA9	CA10	CA11	CA12	CA13		
Activate	ACT	L	L	L	R0	R1	R2	R3	BA0	BA1	BG0	BG1	BG2	CID0	CID1	CID2/DOPIO	11,17,20,21,23,24		
		H	R4	R5	R6	R7	R8	R9	R10	R11	R12	R13	R14	R15	R16	CID3/R17			
RFU	RFU	L	H	L	L	L	L	V	V	V	V	V	V	V	V	V			
		H	V	V	V	V	V	V	V	V	V	V	V	V	V	V			
RFU	RFU	L	H	L	L	L	H	V	V	V	V	V	V	V	V	V			
		H	V	V	V	V	V	V	V	V	V	V	V	V	V	V			
Write Pattern	WRP	L	H	L	L	H	L	H	BA0	BA1	BG0	BG1	BG2	CID0	CID1	CID2/DOPIO	11,15,18,19,20,21,23,24		
		H	V	C3	C4	C5	C6	C7	C8	C9	C10	V	H	H	V	CID3			
Write Pattern w/ Auto Precharge	WRPA	L	H	L	L	H	L	H	BA0	BA1	BG0	BG1	BG2	CID0	CID1	CID2/DOPIO	11,15,18,19,20,21,23,24		
		H	V	C3	C4	C5	C6	C7	C8	C9	C10	V	AP=L	H	V	CID3			
RFU	RFU	L	H	L	L	H	H	V	V	V	V	V	V	V	V	V			
		H	V	V	V	V	V	V	V	V	V	V	V	V	V	V			
Mode Register Write	MRW	L	H	L	H	L	L	MRA0	MRA1	MRA2	MRA3	MRA4	MRA5	MRA6	MRA7	V/DOPIO	8,11,13,20,22,23,24		
		H	OP0	OP1	OP2	OP3	OP4	OP5	OP6	OP7	V	V	CW	V	V	V			
Mode Register Read	MRR	L	H	L	H	L	H	MRA0	MRA1	MRA2	MRA3	MRA4	MRA5	MRA6	MRA7	V/DOPIO	8,13,20,22,23,24,25,26		
		H	L	L	V	V	V	V	V	V	V	V	CW	V	V	V			
Write	WR	L	H	L	H	H	L	BL*=L	BA0	BA1	BG0	BG1	BG2	CID0	CID1	CID2/DOPIO	8,12,15,19,20,21,23,24		
		H	V	C3	C4	C5	C6	C7	C8	C9	C10	V	H	WR Partial=L	V	CID3			
Write w/Auto Precharge	WRA	L	H	L	H	H	L	BL*=L	BA0	BA1	BG0	BG1	BG2	CID0	CID1	CID2/DOPIO	8,12,15,19,20,21,23,24		
		H	V	C3	C4	C5	C6	C7	C8	C9	C10	V	AP=L	WR Partial=L	V	CID3			
Read	RD	L	H	L	H	H	H	BL*=L	BA0	BA1	BG0	BG1	BG2	CID0	CID1	CID2/DOPIO	8,15,19,20,21,23,24		
		H	C2	C3	C4	C5	C6	C7	C8	C9	C10	V	H	V	V	CID3			
Read w/Auto Precharge	RDA	L	H	L	H	H	H	BL*=L	BA0	BA1	BG0	BG1	BG2	CID0	CID1	CID2/DOPIO	8,15,19,20,21,23,24		
		H	C2	C3	C4	C5	C6	C7	C8	C9	C10	V	AP=L	V	V	CID3			

See also, e.g., Ex. CC-89 (Samsung DDR5 RDIMM Datasheet) at 9, 11:

U.S. Patent No. 10,025,731 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use														
	[Table 5] PIN Description <table><tr><th>Pin Name</th><th>Description</th></tr><tr><td>CA[6:0]_A CA[6:0]_B</td><td>SDRAM Address and Command Bus</td></tr><tr><td>CS[1:0]_A CS[1:0]_B</td><td>SDRAM Chip Select</td></tr></table> [Table 7] Input/Output Function Description <table><tr><th>Symbol</th><th>Type</th><th>I/O Levels</th><th>Function</th></tr><tr><td>CS[1:0]_A CS[1:0]_B</td><td>Input</td><td>VDD</td><td>Chip Select : All commands are masked when CS_n is registered HIGH. CS_n provides for external Rank selection on systems with multiple Ranks. CS_n is considered part of the command code. CS_n is also used to enter and exit the parts from power down mode and self refresh mode. While not in self refresh mode the CS_n input buffer operates with the same ODT and VREF parameters as configured by the CA_ODT strap setting or mode register. When in self refresh the CS_n is a CMOS rail to rail signal with DC high and low at 80% and 20% of VDD.</td></tr></table> Below illustrates a DDR5 device’s functional blocks. Although the paper is one by SK hynix, it is believed that Samsung’s DDR5 SDRAMs would include the same functional blocks. Consistent with the command truth table, the functional block shows that the decoded C/A signals are inputs to the “DRAM Core Control,” “Latency Control,” and “WRITE Latency OFFSET Control” blocks, the output of which are provided to additional control blocks to exercise read/write control of the memory core, for example:	Pin Name	Description	CA[6:0]_A CA[6:0]_B	SDRAM Address and Command Bus	CS[1:0]_A CS[1:0]_B	SDRAM Chip Select	Symbol	Type	I/O Levels	Function	CS[1:0]_A CS[1:0]_B	Input	VDD	Chip Select : All commands are masked when CS_n is registered HIGH. CS_n provides for external Rank selection on systems with multiple Ranks. CS_n is considered part of the command code. CS_n is also used to enter and exit the parts from power down mode and self refresh mode. While not in self refresh mode the CS_n input buffer operates with the same ODT and VREF parameters as configured by the CA_ODT strap setting or mode register. When in self refresh the CS_n is a CMOS rail to rail signal with DC high and low at 80% and 20% of VDD.
Pin Name	Description														
CA[6:0]_A CA[6:0]_B	SDRAM Address and Command Bus														
CS[1:0]_A CS[1:0]_B	SDRAM Chip Select														
Symbol	Type	I/O Levels	Function												
CS[1:0]_A CS[1:0]_B	Input	VDD	Chip Select : All commands are masked when CS_n is registered HIGH. CS_n provides for external Rank selection on systems with multiple Ranks. CS_n is considered part of the command code. CS_n is also used to enter and exit the parts from power down mode and self refresh mode. While not in self refresh mode the CS_n input buffer operates with the same ODT and VREF parameters as configured by the CA_ODT strap setting or mode register. When in self refresh the CS_n is a CMOS rail to rail signal with DC high and low at 80% and 20% of VDD.												

Claim 1

Evidence of Use

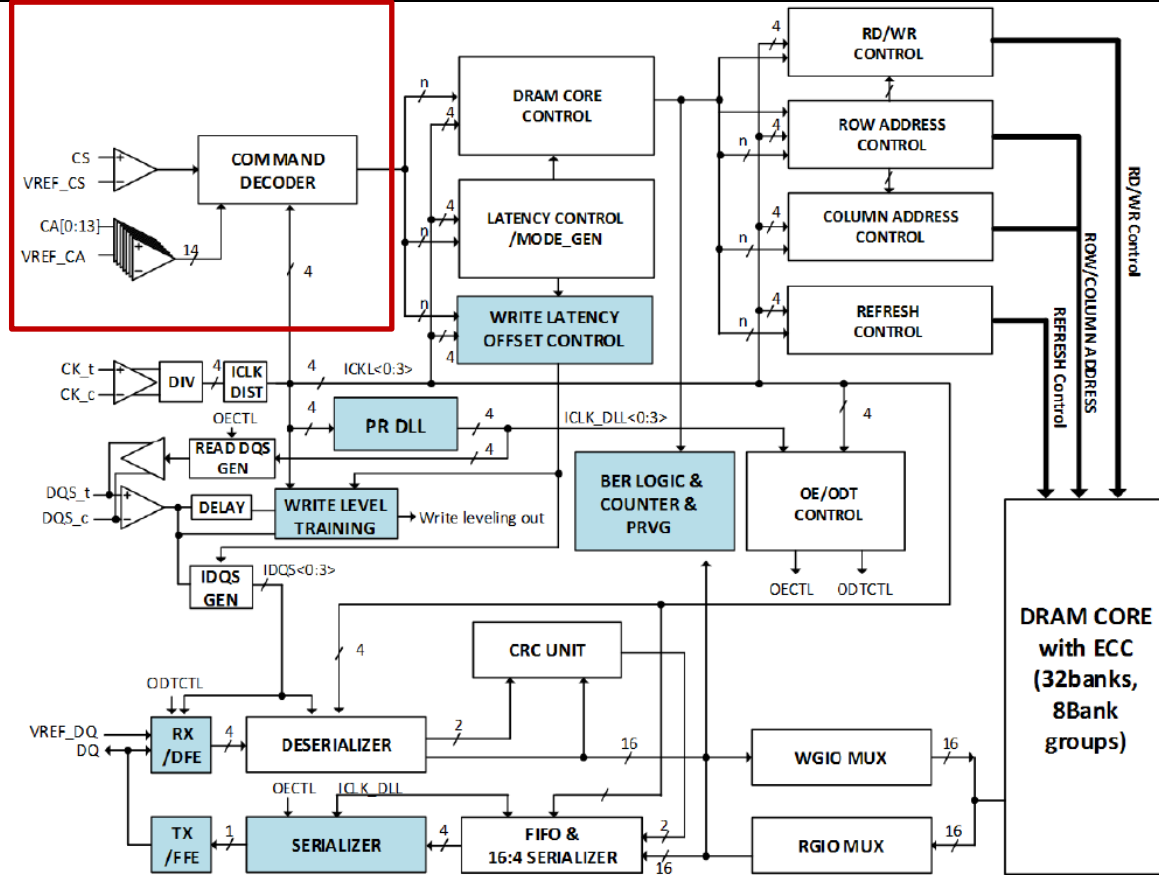


Fig. 2. DDR5 functional block diagram.

Ex. CC-58 (Kim 2020) at 168 (annotated).

U.S. Patent No. 10,025,731 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	This claim element is literally present, as described herein, but is also satisfied under the doctrine of equivalents, and the Samsung Practicing '731 Products are manufactured in a manner that is not substantially different from the requirements of this limitation because they achieve substantially or exactly the same function, in substantially or exactly the same way as set forth in the claim element (as described herein), and achieve substantially or exactly the same result.

Claim 1	Evidence of Use
[1.h] wherein the control circuitry is further configured to dynamically control the at least one programmable impedance matching circuit in the each correction circuit of the set of correction circuits in the each circuit of the at least one circuit based on which of the multiple ranks is selected to communicate data with the system memory controller in	In Samsung's DDR5 RDIMMs and MRDIMMs, <i>control circuitry</i> (such as the RCD, MRCD or the address/control circuit on SDRAMs) <i>is further configured to dynamically control the at least one programmable impedance matching circuit in the each correction circuit of the set of correction circuits in the each circuit of the at least one circuit</i> (e.g., the ODT/RTT circuit identified in 1f) <i>based on which of the multiple ranks is selected to communicate data with the system memory controller in response to the address and control signals</i>. See <i>supra</i> limitation [1.a]-[1.f]. For example, in DDR5 systems, ODT control is encoded into the CS signal received from the memory controller to target a specific rank on the DDR5 DIMMs. By e.g., correctly decoding the input control/address signals received over a 7-bit C/A bus, a single-bit parity bus and a CS bus received over two cycles to output two copies of 14-bit single data rate CA bus outputs and two copies of the chip-select outputs with proper delays obtained through CA/CS training and DFE and directing the two copies of the C/A and C/S outputs to the respective memory devices to control their memory operations,² the RCD together with the command decoding and control circuitry on the SDRAMs (an

² See e.g., Ex. CC-104 (Rambus DDR5 vs DDR4 DRAM) (“[F]or DDR5, the RCD’s CA bus receivers will also require DFE options to ensure good signal reception.”); Ex. CC-72 (Montage M88DR5RCD01) at 1 (“CA, CS and DFE training modes and Dual Frequency support”); *id.* (“CA, CS and DFE training mode[]”; “Dual chip selects”); Ex. CC-74 (Montage M88DR5RCD03) (same); Ex. CC-75 (Montage M88DR5RCD04) (same); Ex. CC-27 (Gen 4 RCD Renesas RRG5004) (“Each channel features a 7-bit double data rate CA bus input, single parity input, two chip-select inputs, and produces two copies of 14-bit single data rate CA bus outputs and chip-select outputs.”); Ex. CC-26 (Gen 3 RCD Renesas RG5R364) at 1 (“Each channel has a 7-bit double data rate CA bus input, a single parity

U.S. Patent No. 10,025,731 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use																										
response to the address and control signals.	example of a control circuitry) controls the operation of the SDRAMs, including affecting the value of the RTT for the specific rank. 3.3 Register Operation 3.3.1 Chip Select Operation The component has two chip select inputs per channel, DCS0_n and DCS1_n, and two copies of each chip select output, QCS0_A_n, QCS1_A_n, QCS0_B_n and QCS1_B_n. Commands are sent to a single rank or multiple ranks, as determined by the DCS[1:0]_n inputs. The number of input chip selects matches the number of output chip selects in each of the two sets (A-outputs and B-outputs). Table 3 — DCS - QxCS Mapping <table><tr><th>Input CS</th><th>Output CS</th></tr><tr><td>DCS0_N_n</td><td>QxCS0_N_n</td></tr><tr><td>DCS1_N_n</td><td>QxCS1_N_n</td></tr></table> 3.3.2 ODT Operation DDR5 does not have a separate ODT (On Die Termination) signal going to the DIMM or the DRAMs. The ODT control is encoded into the chip selects. All commands that utilize the data bus require 2UI on the DCA bus in order to transfer the appropriate number of command and addressing bits. This requirement is used to signal ODT control by how the chip selects are asserted for the two UI. Table 4 — ODT Encoding on the Chip Select <table><tr><th>DCS_n on 1st UI</th><th>DCS_n on 2nd UI</th><th>QxCS[1:0]_n</th><th>DRAM DQ Terminations</th></tr><tr><td>Active</td><td>Inactive</td><td>Target of command</td><td>Read: none (device drives data) Write: RTT_WR</td></tr><tr><td>Active</td><td>Active</td><td>Non-Target Termination should be applied.</td><td>Read: RTT_NOM_RD Write: RTT_NOM_WR</td></tr><tr><td>Inactive</td><td>Inactive</td><td>DRAM ignores commands and the Parked termination will be maintained</td><td>RTT_PARK</td></tr><tr><td>Inactive</td><td>Active</td><td>Illegal</td><td>Illegal</td></tr></table> The host controller will determine the termination at the time of the command, by activating the chip selects appropriately. Ex. CC-32 (JESD82-512) at 30 (annotated).	Input CS	Output CS	DCS0_N_n	QxCS0_N_n	DCS1_N_n	QxCS1_N_n	DCS_n on 1st UI	DCS_n on 2nd UI	QxCS[1:0]_n	DRAM DQ Terminations	Active	Inactive	Target of command	Read: none (device drives data) Write: RTT_WR	Active	Active	Non-Target Termination should be applied.	Read: RTT_NOM_RD Write: RTT_NOM_WR	Inactive	Inactive	DRAM ignores commands and the Parked termination will be maintained	RTT_PARK	Inactive	Active	Illegal	Illegal
Input CS	Output CS																										
DCS0_N_n	QxCS0_N_n																										
DCS1_N_n	QxCS1_N_n																										
DCS_n on 1st UI	DCS_n on 2nd UI	QxCS[1:0]_n	DRAM DQ Terminations																								
Active	Inactive	Target of command	Read: none (device drives data) Write: RTT_WR																								
Active	Active	Non-Target Termination should be applied.	Read: RTT_NOM_RD Write: RTT_NOM_WR																								
Inactive	Inactive	DRAM ignores commands and the Parked termination will be maintained	RTT_PARK																								
Inactive	Active	Illegal	Illegal																								

input, two chip-select inputs, produces two copies of 14-bit single data rate CA bus outputs and two copies of the chip-select outputs.”); Ex. CC-25 (Gen 2 RCD Renesas RG5R256) (same); Ex. CC-24 (Gen 1 RCD Renesas 5RCD0148) (same).

Claim 1

Evidence of Use

Likewise, the C/A and CS circuits on the SDRAMs would decode the incoming C/A and CS signals with proper delay to control the ODT, including the values that it will take:

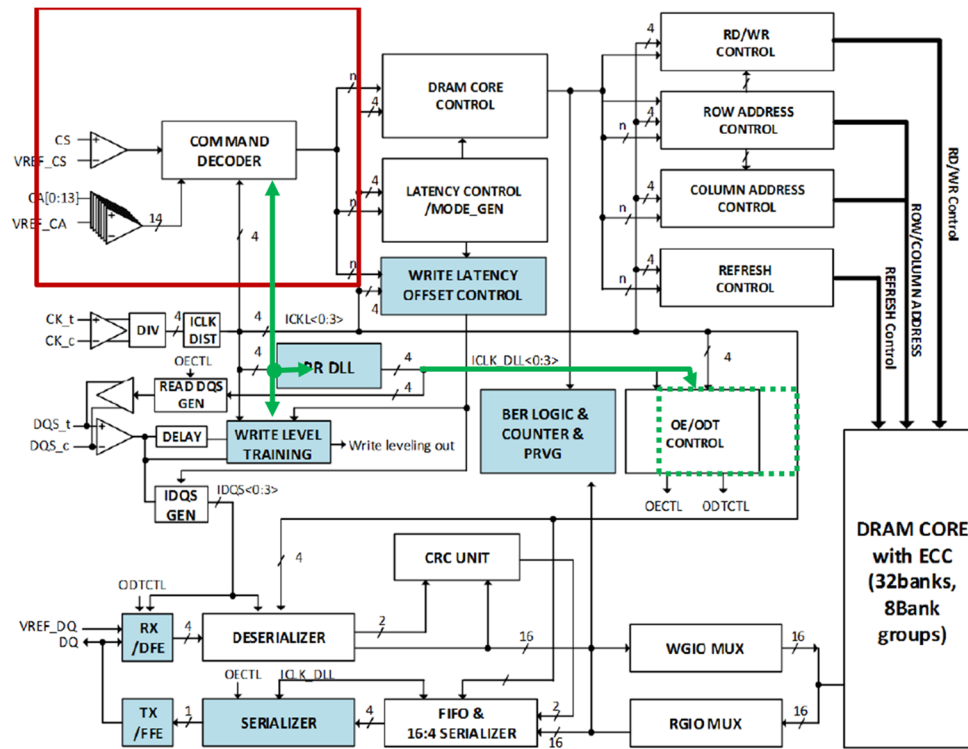
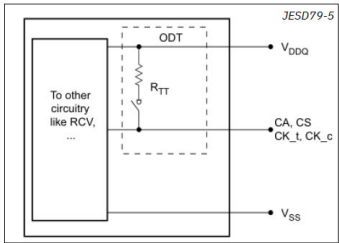


Fig. 2. DDR5 functional block diagram.

Ex. CC-58 (Kim 2020) at 168 (annotated). On information and belief, Samsung SDRAMs would have

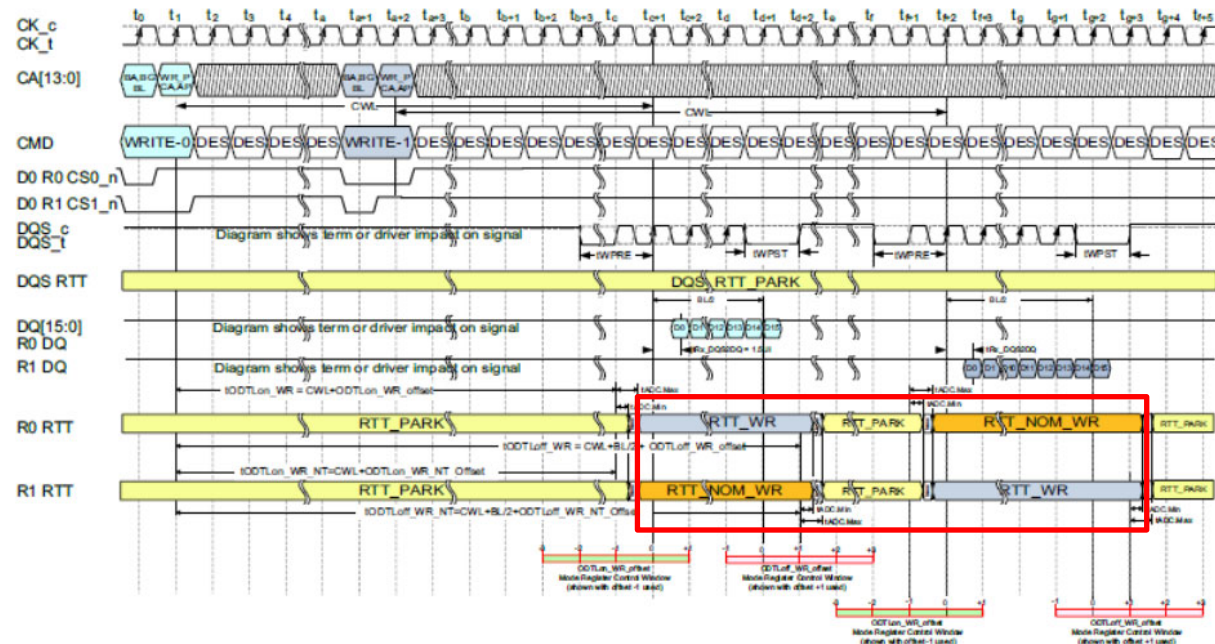
Claim 1	Evidence of Use
	similar basic functional blocks. In DDR5 DIMMs, the dynamic control is exerted by the control circuitry on the DIMMs, and not in response to control dictated by the memory controller. <i>See also</i> Ex. CC-110 (FAC Ex. 7, 25-cv-557 (EDTX)) at 10: On-Die Termination Changes • All ODT is now fully command/register based • No dedicated ODT pin • ODT is now available for CA bus • DQ/DQS still use direct routing from controller to DRAM • CA bus uses fly-by routing • Pin CA_ODT allows different settings for end-of-group • Could use 40Ω for end device but ODT disabled for other devices  For example, in the operation below, the first write operation is to Rank0 and the second to Rank1. As seen in the timing diagram below, the RTT values for Rank 0 (R0) and Rank 1 (R1) differ depending on which of the two ranks is selected for the write operation. The transition from one RTT value to another is dynamic as there are no intermediate MRW operations and depends on the selected rank.

Claim 1

Evidence of Use

JEDEC Standard No. 79-5C_v1.30
Page 323

5.3.3 ODT Timing Diagrams (cont'd)



NOTES:

1. ODTLon_WR, ODTLon_WR_NT, ODTLoFF_WR and ODTLoFF_WR_NT are based on Mode Register settings that can push out or pull in the RTT enable and disable time.
2. The entire range of ODTL control is not shown for simplicity.

Figure 197 — Example of Write to Write Turn Around, Different Ranks

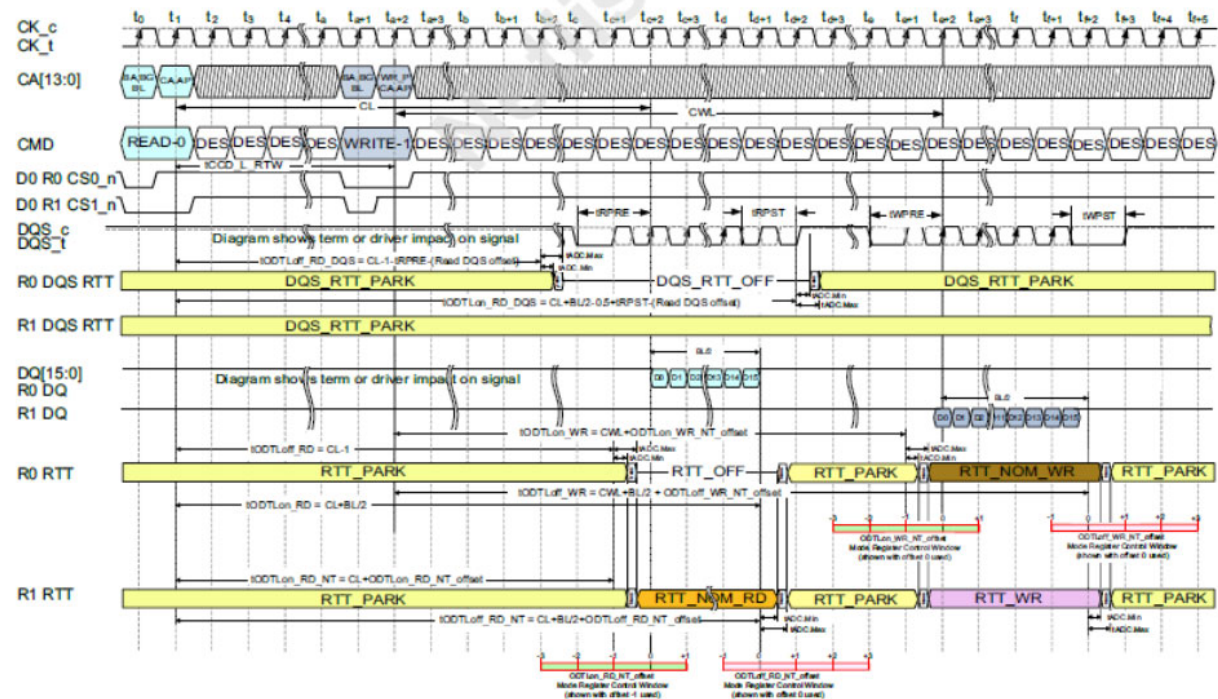
Ex. CC-61 (JESD79-5C_v1.30) at 323 (annotated).

As another example, the timing diagram below shows that in the period when read is operated on Rank0, the RTT for Rank 0 is off, and after the read operation is completed, the RTT for Rank0

Claim 1

Evidence of Use

dynamically transitions to RTT_Park, and then to RTT_NOM_WR when a write operation is performed on Rank1. For Rank1, the RTT value transitions to RTT_NOM_RD during read operation for Rank0, transitions to RTT_Park during the inactive period, and transitions to RTT_WR during the Rank 1 write operation. *See* Ex. CC-61 (JESD79-5C_v1.30) at 325, Fig. 202.

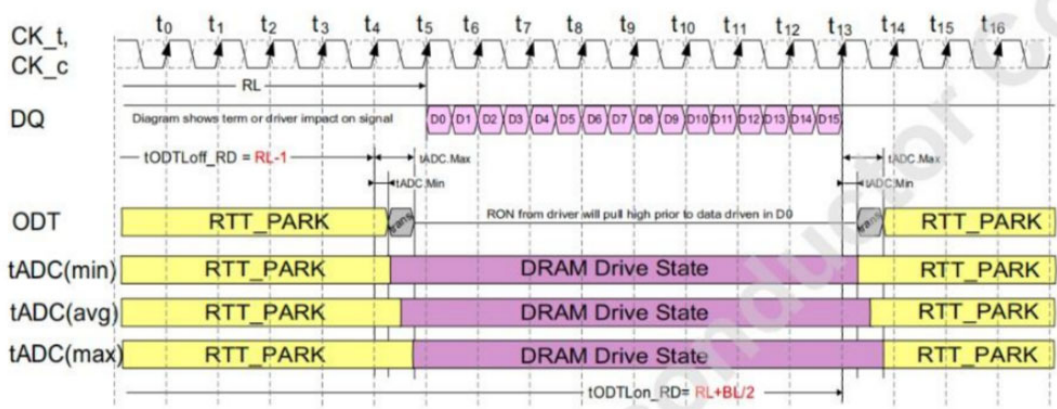


NOTES:

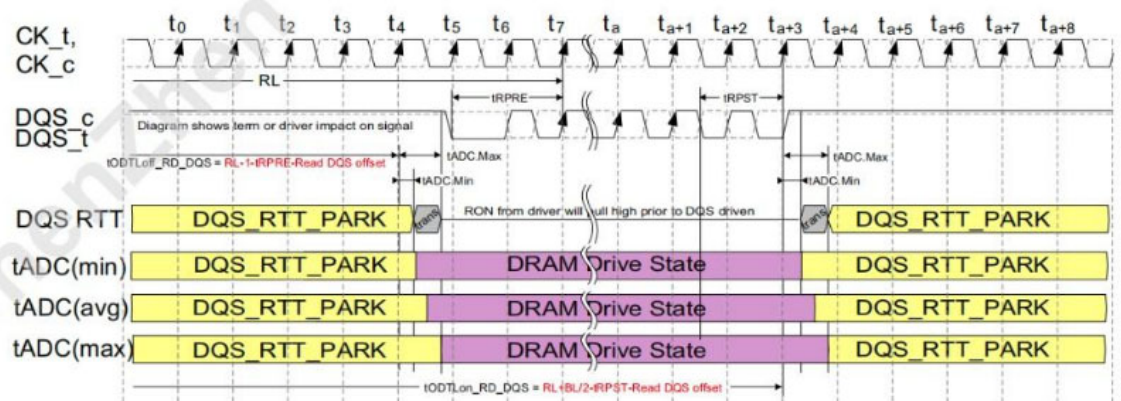
1. ODTLon_WR, ODTLon_WR_NT, ODTLoFF_WR and ODTLoFF_WR_NT are based on Mode Register settings that can push out or pull in the RTT enable and disable time.
2. ODTLon_RD_NT and ODTLoFF_RD_NT are based on Mode Register settings that can push out or pull in the RTT enable and disable time.
3. ODTLon_WR_offset and ODTLoFF_WR_offset not shown for simplicity.
4. Example shown with near ideal timings for termination settings, exact offset configurations will vary based on system designs.

Figure 202 — Example of Read to Write Turn Around, Different Ranks

U.S. Patent No. 10,025,731 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	<i>See also Ex. CC-59 (JEDEC D5 Chapter5) at 7:</i>  NOTE 1 The diagram shows a transition from RTT_PARK to Read DRAM Drive state. When tADC transitions from RTT to Read Drive state, the DRAM RON from the driver will keep the DQ signal high prior to the data driven in D0. No High-Z time during tADC is allowed in this example. NOTE 2 In the case of Term to Write, the host will keep the DQ signal HIGH 4UI prior to the data driven in D0. NOTE 3 The DFE should assume that 4UI prior to D0 the signal is HIGH.

U.S. Patent No. 10,025,731 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	 NOTE 1 The diagram shows a transition from DQS_RTT_PARK to Read DRAM Drive state. When tADC transitions from RTT to Read Drive state, the DRAM RON from the driver will keep the DQS signal high prior to the DQS driven at t5. No High Z time during tADC is allowed in this example. This claim element is literally present, as described herein, but is also satisfied under the doctrine of equivalents, and the Samsung Practicing '731 Products are manufactured in a manner that is not substantially different from the requirements of this limitation because they achieve substantially or exactly the same function, in substantially or exactly the same way as set forth in the claim element (as described herein), and achieve substantially or exactly the same result.